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Personalized ranking for video games based on online reviews: An S-Kano-TOPSIS method integrating requirement categories and public opinion

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ABSTRACT

The rapid expansion of the video game market intensifies customers' difficulty in selecting preference-aligned games. Although online reviews offer valuable insights, effectively leveraging this information remains challenging. To address this, we propose S-Kano-TOPSIS, a personalized ranking method for video games that integrates requirement categories and public opinion. First, BERTopic is used to extract customer requirements (CRs), and their performance is evaluated via sentiment analysis using a BW-CNN model. Then, SHAP is applied to quantify the influence of each CR on customer satisfaction. The Kano model is employed to adjust CR importance based on their influence patterns. Furthermore, to reflect real-world decision-making, we incorporate preference similarity by analyzing reviews of games similar to those the customer has played. Finally, TOPSIS is used to generate rankings tailored to individual needs. Experiments on 72,000 reviews from eight video games demonstrate that the proposed method surpasses baseline approaches across multiple evaluation metrics. These results suggest that S-Kano-TOPSIS offers a structured and quantifiable approach to personalized video game ranking.

1. Introduction

Video games have become one of the most widely consumed forms of entertainment worldwide. In 2023, the industry generated approximately $220 billion in revenue, with projections suggesting this figure could reach $300 billion by 2026 [1]. While the rapid expansion of the video game market provides customers with a diverse range of choices, it also leads to decision fatigue and choice overload [2]. To address this issue, ranking methods have been widely used to help customers efficiently identify games that match their interests [3]. However, existing ranking methods in the gaming industry predominantly rely on revenue or ratings, failing to fully capture the distinct preferences of individual customers and valuable information from online reviews. In other industries, collaborative filtering (CF) and multi-attribute decision-making (MADM) methods are commonly used for ranking [4]. CF, which leverages similarities in customer ratings [5], faces challenges in the video game industry due to the substantial time investment required to experience games, leading to sparse rating data. Meanwhile, MADM-based studies often extract customer preferences from online reviews to rank products [6]. However, these methods primarily focus on group preferences, whereas preferences in the gaming industry are highly individualized and diverse [7]. Consequently, ranking methods centered on group preferences fail to adequately support customers in selecting games that align with their unique preferences. Therefore, it is necessary to propose a personalized ranking method capable of capturing the personalized preferences of video game customers based on online reviews.

Personalized ranking requires accurately measuring how different customers value various aspects of a game, referred to as CRs. Traditional models, such as linear regression, capture CR influence but often oversimplify complex interactions [8]. Machine learning models can capture nonlinear effects but typically operate as black boxes, limiting interpretability . Explainable machine learning addresses this limitation by combining predictive power with interpretability [9]. Among these methods, the Shapley Additive Explanation (SHAP) method is particularly well-suited for quantifying personalized preferences since it can measure the influence of each CR on the model's predicted output for each individual data instance, allowing us to measure how each CR influences satisfaction for every customer [10,11].

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Measuring personalized preferences with SHAP alone may overlook additional adjustment factors. Different CRs can have similar contributions to satisfaction but differ in importance due to asymmetric influence patterns [12] which refer to the phenomenon that improvements in a requirement do not necessarily yield satisfaction gains of the same magnitude as the dissatisfaction caused by its deterioration. The Kano model is a well-established approach for capturing such asymmetries, classifying CRs based on how increases or decreases in their performance affect customer satisfaction [13]. This provides a systematic framework to differentiate CRs that have comparable overall impact but vary in their strategic significance. Furthermore, customer preferences are shaped by public opinion [14]: reviews of previously played or similar games serve as reference points that influence how individuals perceive and prioritize CRs [15]. Neglecting these influences risks misrepresenting true preferences, as social context can systematically bias perceived importance [16]. Therefore, accounting for both asymmetric CR effects and reference-based influences is essential for accurately capturing personalized preferences and generating meaningful rankings.

To address these gaps, this study proposes S-Kano-TOPSIS, a personalized ranking method for video games that integrates requirement categories and public opinion based on online reviews. First, BERTopic extracts CRs, and BW-CNN sentiment analysis transforms them into structured CR-level data. SHAP is then used to capture nonlinear effects and individual-specific preferences. Since SHAP alone cannot reflect asymmetric CR importance, we further incorporate the Kano model to adjust contributions across categories. In addition, reference-based adjustments integrate reviews of similar games, weighting other users' preferences by their similarity to the target customer. Finally, TOPSIS aggregates the adjusted CR preferences to generate a ranking by proximity to the ideal solution. This pipeline ensures that personalized rankings account for nonlinear effects, asymmetric importance, and social influence.

Our research makes contributions along three dimensions:

(1) This study proposes an S-Kano-TOPSIS ranking method, which uniquely combines SHAP, Kano modeling, and reference-based adjustments within TOPSIS. This integration captures nonlinear individual preferences, accounts for asymmetric CR effects, and incorporates public influence—going beyond prior methods that treat these aspects separately.

(2) We develop a review-based method that computes unique personalized preferences for each customer. By leveraging SHAP at the instance level, our approach reveals how different individuals value CRs differently, rather than averaging preferences across groups.

(3) We design an adjustment mechanism that refines raw SHAP-based preferences by incorporating Kano-informed asymmetries and reference-based public opinion. This ensures that the final rankings both preserve individual uniqueness and account for systematic adjustment factors.

The structure of this paper is as follows: Section 2 provides the literature review; in Section 3, an S-Kano-TOPSIS personalized ranking method for video games is proposed. In Section 4, the method proposed in this paper is applied to an experimental study. In Section 5, we discuss the research implications and limitations of our study. Finally, Section 6 presents the findings, the shortcomings, and the future prospects of this paper.

2. Literature review

2.1. Ranking method

Existing ranking methods can broadly be divided into CF-based approaches and MADM-based approaches. CF-based methods infer preferences from historical behavior or user similarity [17,18]. However, they struggle in domains like video games where interaction data are limited [19,20]. Attempts to enhance CF with sentiment analysis or deep learning [21,22] have improved predictive accuracy but still rely heavily on dense behavioral data, limiting their adaptability to dynamic and heterogeneous contexts.

Another commonly used ranking methods are MADM-based ranking methods, a subfield of operations research, systematically evaluate alternatives by jointly considering the relative importance of multiple attributes and aggregating performance information across them [16,23,24]. MADM has been especially effective in ranking applications based on online reviews by leveraging customer preference information embedded in these reviews [25]. Classical techniques such as TOPSIS, VIKOR, and PROMETHEE have been widely applied with various improvements. TOPSIS identifies the option closest to the ideal and farthest from the nadir [26,27]; VIKOR emphasizes compromise solutions by balancing group utility and individual regret [28]; PROMETHEE applies pairwise outranking flows [29]. However, most MADM applications remain group-oriented [7], aggregating preferences across customers and neglecting individual-level heterogeneity.

For personalized ranking, VIKOR's reliance on a regret parameter makes it difficult to set objectively for each customer [30], while PROMETHEE's pairwise comparisons are computationally intensive and do not quantify the distance to an ideal solution [31]. In contrast, TOPSIS directly measures the distance of each alternative to the ideal and nadir points, making it both computationally efficient and naturally adaptable to individualized preference weights. This makes TOPSIS particularly suitable for personalized, review-based game rankings [32].

Building on these insights, this study develops a personalized ranking method for video games based on TOPSIS, which leverages online reviews to model the personalized importance each customer assigns to different CRs. By modeling individual personalized preferences and incorporating the influence of reviews from alternative games, the proposed method aims to deliver more accurate and customer-centered game rankings.

2.2. Preference analysis based on online reviews

With the advancement of internet technology, online reviews have become a key data source for analyzing customer preferences [33–35]. Preference measurement typically begins with customer requirement (CR) analysis, where CRs are extracted from reviews and their performance is evaluated. Advances in natural language processing have provided effective tools for identifying CRs and quantifying their performance levels [33,36–38].

Building on this foundation, customer preferences can then be inferred by examining how CR performance influences satisfaction [9,39]. One stream of traditional research employed linear models to estimate the influence of CRs through regression coefficients [8]. Another stream of research designed rule-based metrics to measure preferences, such as using deviations between CR performance and overall ratings [40] or occurrence frequency as a proxy for preference [41]. In addition, recent studies have applied machine learning models to capture nonlinear relationships between CRs and satisfaction, either by using network parameters [42] or by introducing random perturbations to input features to assess their influence [43].

Linear regression models and rule-based metrics provide interpretability but cannot capture nonlinearities, while machine learning methods such as neural networks and LSTMs can model complex relationships but often operate as black boxes. Moreover, most of these approaches emphasize group-level preference patterns, overlooking individual heterogeneity that is particularly critical in domains such as video games [44,45].

To address these issues, SHAP has been widely adopted across various fields to assess how input features contribute to model outputs [46–49]. Providing more accurate measurements of each CR's performance influence on customer satisfaction by considering individual customer preferences [10]. Rooted in the Shapley value [50] in cooperative game theory, SHAP attributes model predictions to input features by evaluating their marginal contributions across different feature coalitions [51]. This allows it to quantify the individualized influence of each CR on satisfaction while maintaining consistency with the underlying model. Compared with traditional methods, SHAP not only handles nonlinear interactions but also produces preference measures at the level of each customer, making it especially suitable for personalized analysis based on reviews. However, SHAP also has limitations: its additive assumption may oversimplify higher-order dependencies, and equal SHAP values do not necessarily imply equal strategic importance.

To address the asymmetry of CR effects, the Kano model provides a complementary perspective [52]. It categorizes CRs as Attractive feature (A-F), Must-be feature (M-F), One-dimensional feature (O-F) and Indifferent feature (I-F) [53], reflecting how positive or negative changes in performance influence satisfaction [54]. While the original model focused on the presence or absence of CRs [55], more recent extensions consider both positive and negative changes in CR performance and their impact on satisfaction [56]. In the context of online reviews, customer preferences inferred from text can be interpreted as reflecting the effects of CR presence or absence. These inferred preferences can then be used to assign each CR to a Kano category [57].

However, most of the existing studies applying Kano-based classification in decision support typically perform only qualitative analyses of CR importance and do not adjust numerical preference scores according to Kano categories [58]. As a result, they fail to capture the varying strategic importance of different asymmetry patterns—where M-F CRs are most critical, followed by A-F, O-F, and I-F CRs [53,55,57]—even when SHAP or other preference measures indicate similar influence levels. Some studies have further applied Kano categories to assign heuristic weights or initial importance to CRs [59–61], thereby adjusting their overall significance. However, such weights are typically fixed and cannot reflect personalized preference heterogeneity, highlighting the need for more adaptive approaches.

3. Methodology

Fig. 1 illustrates the research framework of the proposed S-Kano-TOPSIS personalized ranking method for video games integrating requirement categories and public opinion, which consists of three main stages.

The first stage is the numeralization of text data, where data collection and preprocessing are followed by topic modeling using the BERTopic model and sentiment analysis using the BW-CNN model, converting the textual data into numerical values. During this stage, CRs are extracted from online reviews, and their performance is assessed, resulting in a structured CR-sentiment matrix. This structured representation provides the foundation for personalized preference measurement in the next stage.

The second stage is customer preference measurement, based on Explainable Machine Learning and the Kano Model. The SHAP method is used to build an explainable machine learning model that measures the influence of each CR on customer satisfaction (review score) as an indication of customer preferences. Additionally, the CRs are classified based on the Kano model, and customer preferences are adjusted according to the CR categories. This stage bridges the raw CR-sentiment data with a nuanced, preference-aware representation that informs the ranking process.

The third stage is the personalized ranking method based on TOPSIS. Since customers' gaming preferences are often influenced by their experiences with previously played and similar games, this stage incorporates review information from other alternative games as a proxy for public opinion to refine preference estimation. By analyzing the preference similarities among customers of different games, the personalized preferences of an individual customer are further adjusted. Finally, the TOPSIS method is applied to generate a personalized ranking list that aligns with the customer's actual needs.

3.1. Numeralization of text data

A prerequisite for personalized ranking based on reviews is to transform unstructured customer feedback into structured numerical representations. Only by doing so can subsequent models consistently measure how specific CRs influence satisfaction. To achieve this, we design a four-step pipeline: First, data collection and preprocessing: ensure that raw review texts are standardized and analyzable. Second, CR extraction using BERTopic: BERTopic is then employed to uncover CR themes from reviews. Third, sentiment analysis with a BW-CNN model: extracting CRs alone is insufficient; it is also necessary to determine customers' sentiment toward them. To this end, we adopt a BW-CNN sentiment model, enabling accurate polarity classification in the gaming domain. Finally, the CR-sentiment matrix links CR mentions with sentiment polarity at the review level, providing a structured input where each entry quantifies both customer attention and evaluation toward a CR. The overall pipeline of this process is illustrated in Fig. 2.

This design ensures that each step feeds directly into the next: preprocessing makes texts tractable, CR extraction identifies the CRs, sentiment analysis attaches evaluative meaning, and the CR-sentiment matrix consolidates these into a structured representation. This pipeline provides the foundation for subsequent individualized preference modeling by transforming unstructured reviews into structured representations.

3.1.1. Data collection and preprocessing

We use Python to make a web crawler to scrape online review data for alternative games from game forums, and preprocess the data using Python. The preprocessing is divided into the following three steps:

(1) **Punctuating.** Online reviews are divided into sub-sentences by punctuations. For the kth review S_{gk} of the gth alternative game P_g, let the set of sub-sentences based on punctuation be $S_{gk} = \{sb_{gk1}, sb_{gk2}, \ldots, sb_{gkL^{gk}}\}$, where sb_{gkl} is the lth sub-sentence in review S_{gk}, and L^{gk} is the number of sub-sentences in review S_{gk}. Here, $g = 1, 2, \ldots, G$, $k = 1, 2, \ldots, K^g$ and $l = 1, 2, \ldots, L^{gk}$. In Chinese, a single word usually consists of 2–4 characters, so sub-sentences shorter than five characters typically contain only one or two words; therefore, they are not split to avoid breaking short phrases that describe the same CR and to preserve semantic integrity for topic modeling and sentiment analysis.

(2) **Stop words removing.** The domain stop-word list is formed by adding proprietary stop-word in the video game industry to the HowNet stop-word list. Based on the domain stop-word list, the stop-words in online reviews are removed using Python.

(3) **Word segmentation and part of speech (POS) tagging.** We use Jieba for word segmentation and POS tagging because it is specifically designed for Chinese text, supports domain-specific dictionaries, and has been widely validated in prior research on Chinese online reviews [62,63]. This ensures accurate tokenization and part-of-speech labeling, which is critical for reliable topic modeling and sentiment analysis.

3.1.2. CR extraction based on bertopic

BERTopic is one of the state-of-the-art topic modeling methods, leveraging BERT (Bidirectional Encoder Representations from Transformers) embeddings to capture rich contextual semantics that traditional methods cannot. This enables a more accurate and meaningful extraction of CRs from review texts for subsequent preference modeling. First, the BERT model converts text data into high-dimensional semantic vectors, capturing contextual semantic information. Second, BERTopic applies UMAP (Uniform Manifold Approximation and Projection) to reduce the dimensionality of the semantic vectors. Third, HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) clusters the dimensionality-reduced semantic vectors.

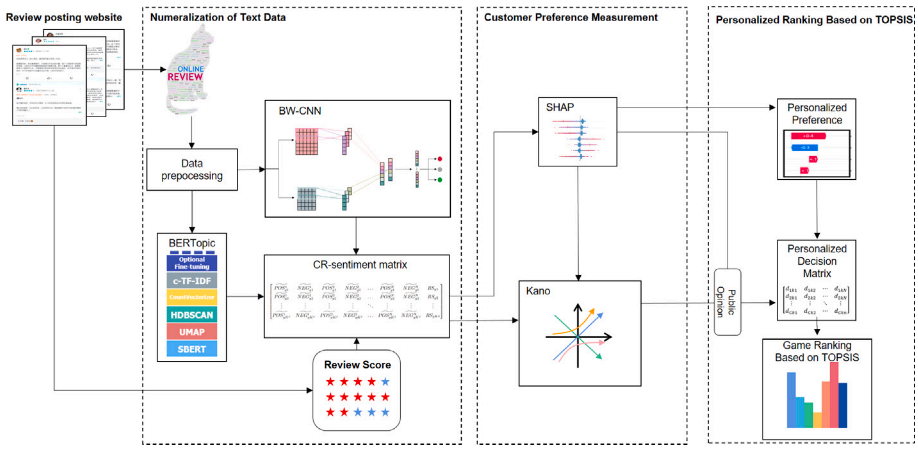

Fig. 1. Framework of S-Kano-TOPSIS personalized ranking method.

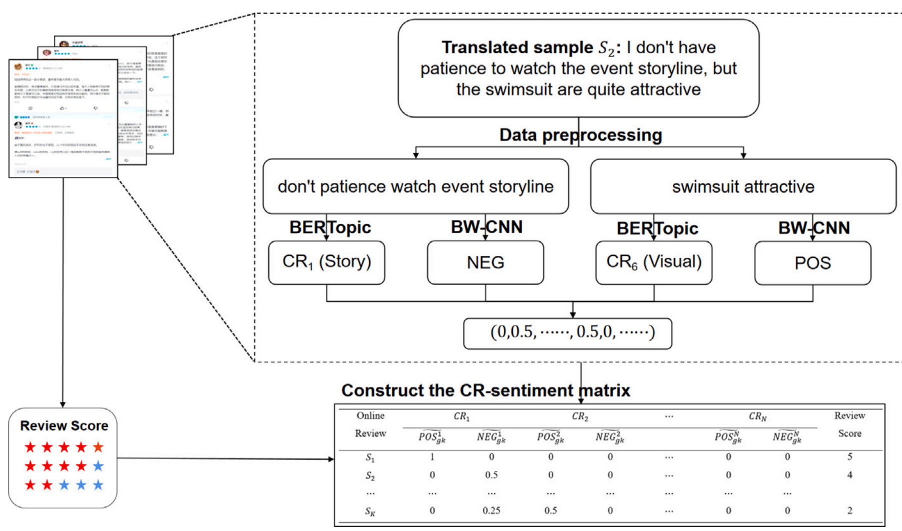

Fig. 2. Pipeline of numeralization of text data.

HDBSCAN is a density-based clustering algorithm capable of automatically determining the number of clusters and detecting outliers (noise). Finally, BERTopic generates topics by extracting keywords from each cluster. These topics represent customers' core needs and concerns as expressed in reviews. Experts can identify and summarize these topics to extract CRs from online reviews.

To ensure the consistency and generalizability of the CR definitions across games, we conducted a structured consolidation process. To ensure the consistency and generalizability of CR definitions across games, we conducted a structured consolidation process with a panel of four experts (two industry practitioners and two doctoral students). Initially, no CR were predefined. Each candidate topic was sequentially examined: if it semantically matched an existing CR, it was assigned to that CR; if not, a new CR was created. Through iterative discussion and majority agreement, the expert panel progressively refined the set of CRs. Although traditional inter-rater reliability metrics such as Fleiss' Kappa cannot be directly applied to this incremental, open-ended classification, we conducted two additional review sessions after the initial consolidation to verify and refine the category assignments. This process ensured a representative and consistent set of CRs for subsequent analysis.

After the above steps, any review sub-sentence sb_{gkl} can be transformed into the corresponding CR vector $V_{gkl} = (CR_{gkl1}, CR_{gkl2}, \dots, CR_{gklN})$, where N is the number of CRs. If the sub-sentence sb_{gkl} belongs to the nth CR, then $CR_{gkln} = 1$; otherwise, $CR_{gkln} = 0$.

Considering that the CRs extracted from different alternative game reviews may differ, we classify the CRs into two types based on their frequency of occurrence in the reviews of different alternative games: basic CRs and unique CRs. Some feature selection and aggregation methods regard the 50% threshold as a reasonable criterion for identifying main features [64,65]. Accordingly, a basic CR refers to a CR that appears in more than half of the alternative games, while a unique CR refers to a CR that appears in half or fewer of the alternative games.

3.1.3. Sentiment analysis based on BW-CNN

To determine customers' sentiment toward each CR, we adopt a dual-channel BW-CNN model, which has been demonstrated [9]. This model combines general language embeddings (BERT) and domain-specific embeddings (Word2Vec) to capture both broad linguistic patterns and video game-specific expressions. In addition, it integrates embeddings at both the Chinese character level and the word level, allowing the model to better capture semantic relationships within and across words—a critical advantage given the complex relationships between Chinese characters and words, as well as the unique language habits often found in online video game reviews. These features make BW-CNN particularly suitable for accurate sentiment classification in the gaming domain and for subsequent individualized preference modeling.

The model treats sentiment analysis as a ternary classification task, labeling each sub-sentence as positive, negative, or neutral.

(1) **Double Embedding Strategy.** To leverage both general language understanding and domain-specific nuances, two embeddings are employed in parallel:

- **BERT:** Provides 768-dimensional embeddings at the Chinese character level, benefiting from rich pretraining but limited in domain-specific terms.
- **Word2Vec:** Trained on video game reviews, offers 300-dimensional word-level vectors, capturing unique expressions specific to the domain.

These two embedding matrices ($sb_{gkl\text{-}BERT}$ and $sb_{gkl\text{-}Word2Vec}$) represent each sub-sentence and serve as inputs to two separate CNN channels.

(2) **Dual-Channel Convolution and Pooling.** Each embedding channel is processed by a one-dimensional convolutional layer followed by a max-pooling layer. Multiple filter heights $h \in \{2, 3, 4, \dots\}$ are used to capture h-gram features of varying lengths. For each filter height, $q = 4$ convolutional filters with widths equal to the embedding dimension are applied. Given an input embedding matrix $sb_{gkl} \in \mathbb{R}^{L\times d}$ of L tokens (rows) and d-dimensional embeddings (columns), a convolution filter slides vertically over the matrix with a window size of h. At each position $\beta \in \{1, \dots, L-h+1\}$, the feature is computed as:

$$p_\beta = f\left(\omega \cdot sb_{gkl}[\beta : \beta + h - 1] + b\right) \tag{1}$$

where $\omega \in \mathbb{R}^{h\times d}$ is the filter weight matrix, b is a bias term, $f(\cdot)$ is the ReLU activation function, and $sb_{gkl}[\beta : \beta + h - 1]$ denotes a window of h consecutive rows (tokens) from the input. After convolution, max-pooling is applied over each resulting feature map to retain the most salient signal. The pooled features from all filters and filter heights are concatenated. Finally, the outputs from both the BERT and Word2Vec channels are concatenated to form the final feature vector.

(3) **Classification.** The combined feature vector is passed through a fully connected layer and a Softmax classifier to yield the sentiment probabilities $\{\theta_{POS}, \theta_{NEU}, \theta_{NEG}\}$. The sentiment label is determined by the highest probability.

This hybrid architecture enhances both general linguistic understanding and domain adaptation, resulting in more accurate sentiment classification of CRs in video game reviews.

3.1.4. Construction of the CR-sentiment matrix

The frequency of mentions of different CRs in online reviews represents the customer's attention to each CR. In this section, we consider customer attention and construct the CR-sentiment matrix. The specific construction process is as follows:

Based on the CR vectors constructed in Section 3.1.2 and the sentence sentiment polarity obtained in Section 3.1.3, each sub-sentence sb_{gkl} can be represented by a vector $(POS^1_{gkl}, NEG^1_{gkl}, POS^2_{gkl}, NEG^2_{gkl}, \dots, POS^N_{gkl}, NEG^N_{gkl})$.

For each sub-sentence sb_{gkl} and the nth CR:

(1) If $CR_{gkln} = 1$ and the sentiment polarity of sub-sentence sb_{gkl} is positive, then $POS^n_{gkl} = 1$ and $NEG^n_{gkl} = 0$;

(2) If $CR_{gkln} = 1$ and the sentiment polarity of sub-sentence sb_{gkl} is negative, then $POS^n_{gkl} = 0$ and $NEG^n_{gkl} = 1$;

(3) If $CR_{gkln} = 0$ or the sentiment polarity of sub-sentence sb_{gkl} is neutral, then $POS^n_{gkl} = 0$ and $NEG^n_{gkl} = 0$.

For each online review S_{gk}, $POS^n_{gk} = \sum_l POS^n_{gkl}$, $NEG^n_{gk} = \sum_l NEG^n_{gkl}$.

Furthermore, since the number of sub-sentences varies across each review, and this difference can interfere with the machine learning model's measurement of the relationship between CRs and customer satisfaction, it is necessary to normalize each review according to Eqs. (2)–(3). Specifically, instead of using absolute counts, we use proportions. $\widetilde{POS^n_{gk}}$ and $\widetilde{NEG^n_{gk}}$ represent the normalized positive and negative sentiment values for CR$_n$ in P_g, respectively. These values represent the ratio of the number of positive or negative sub-sentences related to CR$_n$ to the total number of positive and negative sub-sentences across all CRs in the review. With this normalization, the sum of all CR sentiment values in a review is fixed to 1, ensuring comparability across reviews of different lengths. This measure is referred to as the customer attention-based sentiment intensity index (SII). The SII can be regarded as the performance of CR.

$$\widetilde{POS^n_{gk}} = \frac{POS^n_{gk}}{\sum_{u=1}^{N_g} POS^u_{gk} + \sum_{u=1}^{N_g} NEG^u_{gk}} \tag{2}$$

$$\widetilde{NEG^n_{gk}} = \frac{NEG^n_{gk}}{\sum_{u=1}^{N_g} POS^u_{gk} + \sum_{u=1}^{N_g} NEG^u_{gk}} \tag{3}$$

where N_g is the number of CRs in P_g.

Furthermore, based on the sentiment intensity index, we construct the CR-sentiment matrix of P_g as Eq. (4) is given in Box I. Where RS_{gk} is the review score for each online review that can be crawled from the online review forum. The review score represents the overall evaluation of the game by the customer who posted the review, and it can be considered as the customer's satisfaction with the game.

3.2. Customer preference measurement

In the personalized ranking process, measuring customer preferences plays a critical role in understanding the importance and influence of each CR. As shown in Fig. 3, this section consists of two main components: First, interpretable machine learning model: SHAP is applied to estimate, for each customer, how changes in CR performance shift satisfaction, producing quantitative and review-level influence values. Second, preference measurement based on SHAP values and Kano model: raw SHAP influences alone cannot capture asymmetric patterns. The Kano model complements SHAP by categorizing CRs based on the balance of positive versus negative responses. This categorization provides adjustment rules, ensuring that CRs with the same SHAP values but different asymmetry patterns are weighted appropriately.

Together, SHAP offers personalized quantitative measures, while Kano introduces asymmetry-aware adjustments. This combined process transforms structured CR signals into preference weights that are both individualized and strategy-sensitive, laying the foundation for reliable personalized ranking.

3.2.1. Interpretable machine learning model

In order to improve the model interpretability without compromising the model's fit, this study uses the SHAP method to construct an interpretable machine learning model. This method can uncover the influence of each CR's performance on the review score. This study aims to establish a Back Propagation Neural Network (BPNN) model to learn the relationship between each $\widetilde{POS^n_{gk}}$ ($\widetilde{NEG^n_{gk}}$) and RS_{gk}.

Using the data from the CR-sentiment matrix, we train the BPNN model to obtain the trained model md. By using md and a given set of v-dimensional input variables $X = [x_1, \dots, x_v]$, SHAP values can be calculated to explain the contribution of each feature value in this set to the output value. The SHAP value of x_δ, denoted as $\phi_\delta(md, X)$, can be calculated using Eq. (5):

$$\phi_\delta(md, X) = \sum_{Z\subseteq X} \frac{|Z|!(v - |Z| - 1)!}{v!} \left[md(Z) - md(Z \setminus \delta)\right] \tag{5}$$

where Z is a subset of all the input features of the model, and $|Z|$ represents the size of Z.

Based on the interpretable machine learning model, the positive and negative SIIs' influence on the review score can be measured. The influence is denoted as ϕ^{POS}_{gkn} and ϕ^{NEG}_{gkn}, respectively.

$$\begin{bmatrix} \widetilde{POS}^1_{g1} & \widetilde{NEG}^1_{g1} & \widetilde{POS}^2_{g1} & \widetilde{NEG}^2_{g1} & \cdots & \widetilde{POS}^N_{g1} & \widetilde{NEG}^N_{g1} & RS_{g1} \\ \widetilde{POS}^1_{g2} & \widetilde{NEG}^1_{g2} & \widetilde{POS}^2_{g2} & \widetilde{NEG}^2_{g2} & \cdots & \widetilde{POS}^N_{g2} & \widetilde{NEG}^N_{g2} & RS_{g2} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \widetilde{POS}^1_{gK^g} & \widetilde{NEG}^1_{gK^g} & \widetilde{POS}^2_{gK^g} & \widetilde{NEG}^2_{gK^g} & \cdots & \widetilde{POS}^N_{gK^g} & \widetilde{NEG}^N_{gK^g} & RS_{gK^g} \end{bmatrix} \tag{4}$$

Box I.

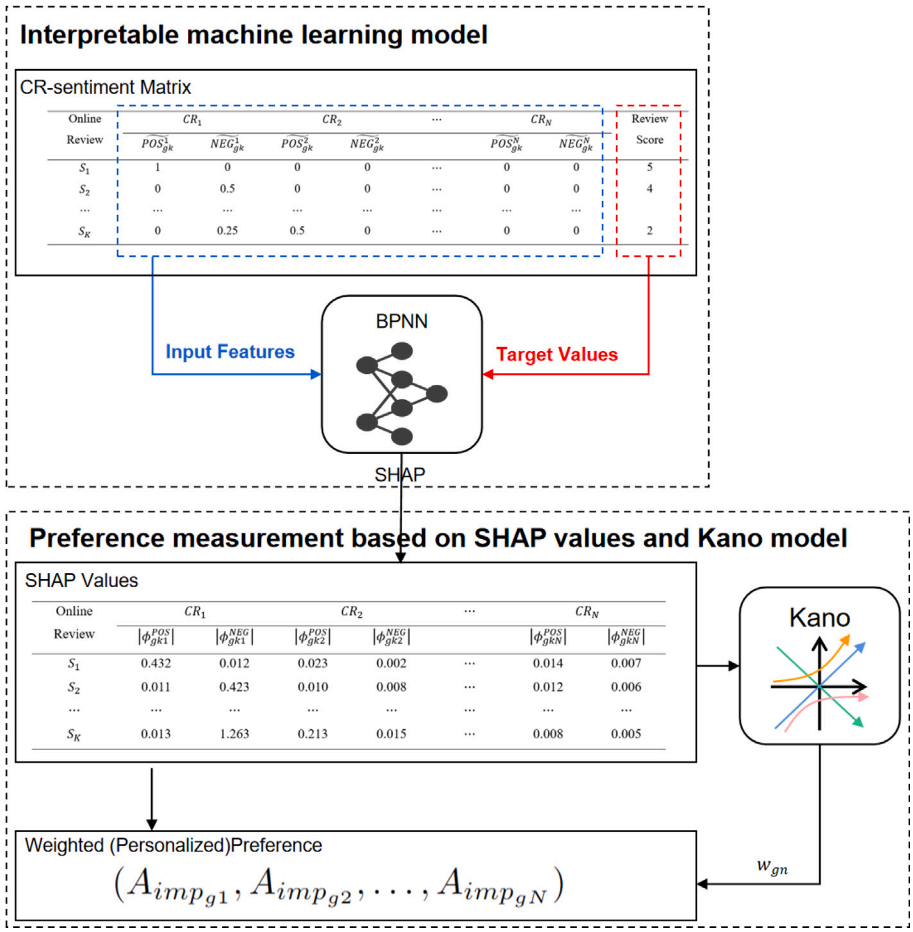

Fig. 3. Pipeline of customer preference measurement.

3.2.2. *Preference measurement based on SHAP values and kano model*

Based on the SHAP values, the influence of each SII on the review score can be measured. The absolute average of the SHAP values is calculated using Eqs. (6)–(7):

$$\left|\overline{\phi^{POS}_{gn}}\right| = \frac{1}{K^g}\sum_{k=1}^{K^g}\left|\phi^{POS}_{gkn}\right| \tag{6}$$

$$\left|\overline{\phi^{NEG}_{gn}}\right| = \frac{1}{K^g}\sum_{k=1}^{K^g}\left|\phi^{NEG}_{gkn}\right| \tag{7}$$

Missing values for unmentioned basic CRs need to be imputed to ensure a complete and comparable preference matrix. Common imputation methods include zero imputation, mean imputation, median imputation, minimum imputation, KNN imputation, and multiple imputation [66]. Since zero imputation may underestimate the subtle attention a customer might implicitly have toward a CR; mean or median imputation would overestimate the importance of CRs that were not mentioned by the customer, while KNN or multiple imputation may be unreliable when no prior observations exist for a given CR. We use the minimum values $\min_{(g,n)}\left|\overline{\phi^{POS}_{gn}}\right|$ and $\min_{(g,n)}\left|\overline{\phi^{NEG}_{gn}}\right|$. Because these unmentioned CRs have relatively low attention and influence among the corresponding customers.

The average influence of CR_n on P_g can be denoted as $Imp_{gn} = |\overline{\phi^{POS}_{gn}}| + |\overline{\phi^{NEG}_{gn}}|$. However, even if two different Kano categories of CR have the same magnitude of influence, they contribute to satisfaction in distinct ways.

Traditional Kano classification relies on two-dimensional functional /dysfunctional questionnaires, which are not feasible for large-scale online review data. A large body of research [12,54,56,57,67,68] has adopted the approach of using positive and negative influences of product features on customer satisfaction to represent the functional/dysfunctional dimensions, typically using means or medians of these measures as thresholds for Kano classification [12,56,57,67]. Following this rationale, we define the Kano classification rules to categorize the CRs as Eq. (8).

$$\begin{cases} \mathrm{CR}_{gn} \in \text{O-F}, & \text{if } |\overline{\phi_{gn}^{POS}}| \geq MED_{POS} \text{ and } |\overline{\phi_{gn}^{NEG}}| \geq MED_{NEG}, \\ \mathrm{CR}_{gn} \in \text{A-F}, & \text{if } |\overline{\phi_{gn}^{POS}}| \geq MED_{POS} \text{ and } |\overline{\phi_{gn}^{NEG}}| < MED_{NEG}, \\ \mathrm{CR}_{gn} \in \text{M-F}, & \text{if } |\overline{\phi_{gn}^{POS}}| < MED_{POS} \text{ and } |\overline{\phi_{gn}^{NEG}}| \geq MED_{NEG}, \\ \mathrm{CR}_{gn} \in \text{I-F}, & \text{if } |\overline{\phi_{gn}^{POS}}| < MED_{POS} \text{ and } |\overline{\phi_{gn}^{NEG}}| < MED_{NEG}. \end{cases} \tag{8}$$

Where $MED_{POS} = \text{median}(|\overline{\phi_{gn}^{POS}}|)$, $MED_{NEG} = \text{median}(|\overline{\phi_{gn}^{NEG}}|)$.

We assign different weights w_{gn} to each CR_{gn} based on its Kano category. Prior studies have often relied on fixed or heuristic weights to distinguish Kano categories [59–61]. While effective in reflecting categorical differences, such static settings fail to capture individual-level asymmetry in customer responses. According to the Kano model, CRs of different classifications have different levels of importance. M-F CRs directly affect the basic satisfaction, and their quality deterioration significantly lowers the customer experience, so they are assigned the highest weight. O-F CRs have a clear difference between satisfaction and dissatisfaction, so they are assigned the second-highest weight. A-F CRs bring surprise, but their quality deterioration does not lead to dissatisfaction; hence, they have lower importance. I-F CRs have minimal influence on satisfaction and are assigned the lowest importance.

To improve flexibility and align with the asymmetric logic of Kano theory, we introduce a SHAP-based adjustment mechanism. Specifically, M-F CRs are given higher weights when their negative influence dominates, while A-F CRs are down-weighted when their positive influence dominates. This asymmetry is quantified through normalized differences of positive vs. negative SHAP values, δ_g^M and δ_g^A, defined in Eqs. (10)–(11). The use of normalized differences reflects the core principle of Kano asymmetry: the relative dominance of dissatisfaction vs. satisfaction determines a feature's criticality. By taking the average difference across CRs within each category, the adjustment is stabilized and avoids bias from outliers, while the normalization ensures comparability across customers and CRs. Since δ_g^M and δ_g^A are bounded within $[0, 1]$, a scaling factor of $1/2$ is applied to control the boundary of adjusted weights. This design guarantees that w_{gn} falls within $[0, 1.5]$, highlighting M-F CRs, moderately down-weighting A-F CRs, and keeping I-F CRs the least important but non-negative. The resulting weights are shown in Eq. (9).

$$w_{gn} = \begin{cases} w_g^{O-F} = 1, \\ w_g^{M-F} = 1 + \frac{\delta_g^M}{2}, \\ w_g^{A-F} = 1 - \frac{\delta_g^A}{2}, \\ w_g^{I-F} = 1 - \frac{\delta_g^M + \delta_g^A}{2}. \end{cases} \tag{9}$$

$$\delta_g^M = \max\left(0, \frac{\overline{|\phi_{gn}^{NEG}|}_{M-F} - \overline{|\phi_{gn}^{POS}|}_{M-F}}{\overline{|\phi_{gn}^{NEG}|}_{M-F} + \overline{|\phi_{gn}^{POS}|}_{M-F} + \epsilon}\right) \tag{10}$$

$$\delta_g^A = \max\left(0, \frac{\overline{|\phi_{gn}^{POS}|}_{A-F} - \overline{|\phi_{gn}^{NEG}|}_{A-F}}{\overline{|\phi_{gn}^{POS}|}_{A-F} + \overline{|\phi_{gn}^{NEG}|}_{A-F} + \epsilon}\right) \tag{11}$$

Here, $\overline{|\phi_{gn}^{POS}|}_{M-F}$ and $\overline{|\phi_{gn}^{NEG}|}_{M-F}$ denote the average positive and negative SHAP values of CRs in the M-F category, and similarly for A-F. A small constant ϵ is added to avoid division by zero.

Thus, the final customer preference for the nth CR after Kano adjustment for P_g is $A_{imp_{gn}} = w_{gn} \times Imp_{gn}$. The customer preference vector for P_g can be represented as $Pf_g = (A_{imp_{g1}}, A_{imp_{g2}}, \ldots, A_{imp_{gN}})$

It should be noted that the variables $|\overline{\phi_{gn}^{POS}}|$, $|\overline{\phi_{gn}^{NEG}}|$, and the corresponding threshold values MED_{POS} and MED_{NEG} in Eqs. (6)–(8), as well as the Kano category assignment in Eq. (8) and the Kano-based weight computation in Eq. (9), can be calculated either at the game level or at the individual customer level.

3.3. Personalized ranking method based on TOPSIS

In this study, we use the TOPSIS method to perform a personalized ranking for any customer u. As shown in Fig. 4, the personalized ranking pipeline based on TOPSIS leads to a final ranking score that better reflects individual customer preferences. The specific steps are as follows:

(1) **Calculation of CRs' performance:** The performance SEN_g^n of CR_{gn} can be represented by the proportion of positive reviews, as Eq. (12):

$$SEN_g^n = \frac{\sum_{k=1}^{K^g} \widetilde{\mathrm{POS}}_{gk}^n}{\sum_{k=1}^{K^g} \widetilde{\mathrm{POS}}_{gk}^n + \sum_{k=1}^{K^g} \widetilde{\mathrm{NEG}}_{gk}^n} \tag{12}$$

(2) **Handling missing values:** Since each game corresponds to a single data point, sophisticated imputation methods such as KNN are not applicable due to the limited data. For basic CRs, which exist objectively across most games, an unmentioned CR does not imply poor performance but merely reflects a lack of attention. Imputing missing values with the minimum or zero would underestimate their true performance; therefore, missing data is imputed by the average performance of that CR across all games. In contrast, unique CRs appear only in a subset of games. Using the mean for imputation would reduce their distinctiveness, weakening the advantage of games containing these features, while zero-value imputation would unfairly lower their overall TOPSIS ranking. Hence, missing values in games without these CRs are filled using the minimum performance value of that CR across all games.

(3) **Adjusted personalized decision matrix considering public opinion:** For any customer u, the personalized importance of a CR can be obtained in two cases. If customer u has explicitly mentioned CR_n, its importance is computed as

$$Rimp_{un} = w_{un} \times (|\phi_{un}^{POS}| + |\phi_{un}^{NEG}|),$$

where w_{un} is the Kano weight of CR_n for customer u, and ϕ_{un}^{POS} and ϕ_{un}^{NEG} are the positive and negative SHAP values of the CR. If CR_n is not explicitly mentioned by u, its importance is inferred in two steps. First, we calculate the $Aimp_{gn}^u = \frac{1}{|G_u|}\sum_{g \in G_u} Aimp_{gn}$, the average of $Aimp_{gn}$ of games mentioned by customer u. Where G_u is the set of games reviewed by customer u, and $Aimp_{gn}$ is the average SHAP-based importance of CR_n across reviewers of game g. This represents how much CR_n typically matters in the games that u has engaged with. Second, to account for the effect of public opinion, we refine this preference by considering other customers with similar preference structures. Specifically, the adjustment is defined as $Rimp_{un} = (1 - \alpha)\, Aimp_{gn}^u + \alpha \cdot \frac{\sum_h (sim(Pf_h, Pf_u) \cdot Aimp_{hn})}{\sum_h sim(Pf_h, Pf_u)}$,. Where Pf_u is the average preference profile of games in G_u, Pf_h is the preference profile of another customer h, and $sim(\cdot,\cdot)$ is the cosine similarity. This adjustment serves as a proxy for public opinion because it selectively incorporates insights from customers whose preference structures resemble those of u, rather than aggregating indiscriminately across all users. By weighting peers according to similarity, it preserves individualized preferences while still leveraging broader feedback, avoiding simple majority reinforcement. Using cosine similarity of preference vectors as a measure of customer similarity has been widely adopted in personalized ranking and recommendation research [69,70].
Moreover, we distinguish between basic and unique CRs in this adjustment. For basic CRs, which are common across many games, the public opinion adjustment ensures that individual rankings remain aligned with these widely recognized standards. For unique CRs, however, the adjustment is not applied,

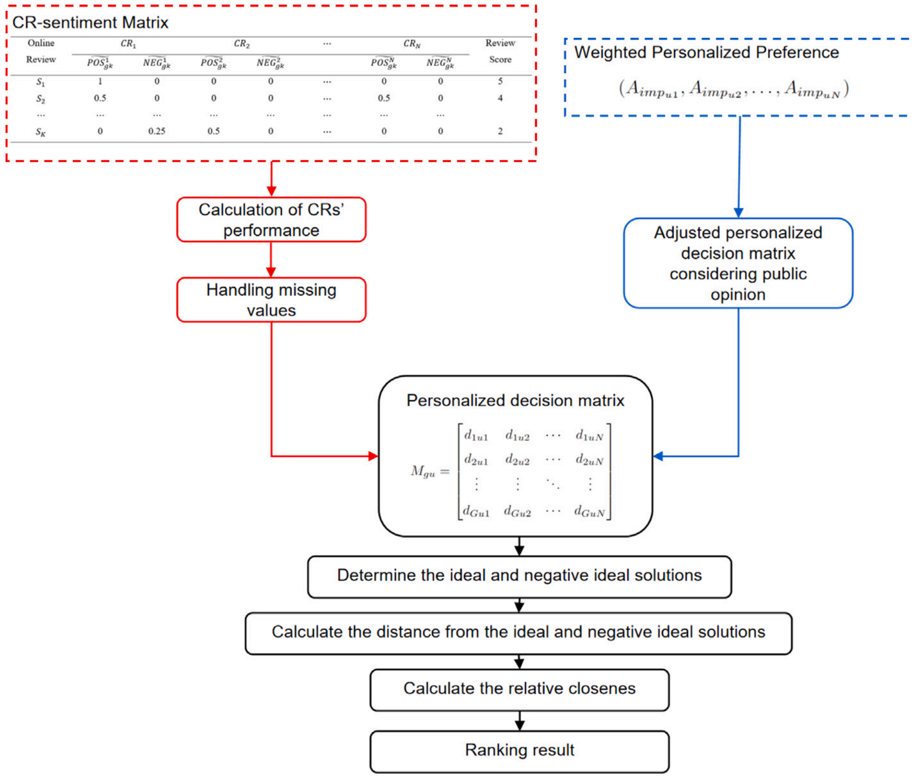

Fig. 4. Pipeline of personalized ranking method based on TOPSIS.

i.e., $Rimp_{un} = Aimp^{u}_{gn}$, since their value lies in personalization and niche differentiation rather than broad consensus [71].
Based on the performance SEN^n_g of CR_{gn} and the customer's personalized preference, the personalized decision matrix for customer u of game P_g, after normalization, can be expressed as Eq. (13):

$$M_{gu} = \begin{bmatrix} d_{1u1} & d_{1u2} & \cdots & d_{1uN} \\ d_{2u1} & d_{2u2} & \cdots & d_{2uN} \\ \vdots & \vdots & \ddots & \vdots \\ d_{Gu1} & d_{Gu2} & \cdots & d_{GuN} \end{bmatrix} \tag{13}$$

where $d_{gun} = \frac{SEN^n_g}{\sqrt{\sum_g (SEN^n_g)^2}} \times \frac{Rimp_{un}}{\sum_n Rimp_{un}}$.

(4) **Determine the ideal and negative ideal solutions:** Calculate the ideal solution A^+_{un} and negative ideal solution A^-_{un} for each CR. The ideal solution is the maximum value of each CR in the personalized decision matrix, while the negative ideal solution is the minimum value of each CR.

$$A^+_{un} = \max_g(d_{gun}) \tag{14}$$

$$A^-_{un} = \min_g(d_{gun}) \tag{15}$$

(5) **Calculate the distance from the ideal and negative ideal solutions:** Calculate the Euclidean distance from each game in the weighted normalized matrix to the ideal and negative ideal solutions as Eqs. (16)–(17)

$$D^+_{gu} = \sqrt{\sum_{n=1}^{N} (d_{gun} - A^+_{un})^2} \tag{16}$$

$$D^-_{gu} = \sqrt{\sum_{n=1}^{N} (d_{gun} - A^-_{un})^2} \tag{17}$$

(6) **Calculate the relative closeness:** Based on the distances to the ideal and negative ideal solutions, calculate the relative closeness of each game as Eq. (18):

$$C_{gu} = \frac{D^-_{gu}}{D^+_{gu} + D^-_{gu}} \tag{18}$$

where C_{gu} represents the relative closeness of the overall performance of P_g to the ideal solution for the uth customer. The larger the value, the closer the game's overall performance is to the ideal solution, and the higher its ranking.

Through the above steps, the personalized preferences of customers can be adjusted based on CR categories and public opinion, and a personalized game ranking can be provided for each customer using the TOPSIS method.

To evaluate the effectiveness and reliability of the proposed personalized ranking method, we employ *Spearman's rank correlation coefficient* (ρ) to measure the consistency between the predicted game rankings and the actual customer scores. As a non-parametric measure of statistical dependence, Spearman's ρ assesses how well the relationship between two rankings can be described using a monotonic function. A higher correlation indicates a better alignment between the personalized ranking results and customer preferences.

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)} \tag{19}$$

where ρ denotes the Spearman's rank correlation coefficient, n is the number of games, and d_i is the difference between the predicted and actual ranks of the ith game. The coefficient ρ ranges from −1 to 1, where a value closer to 1 indicates stronger agreement between the two rankings.

4. Experiment results

In this section, to validate the effectiveness and feasibility of the proposed method in the video game industry, we focus on a single category of games—romance or otome games—because games within

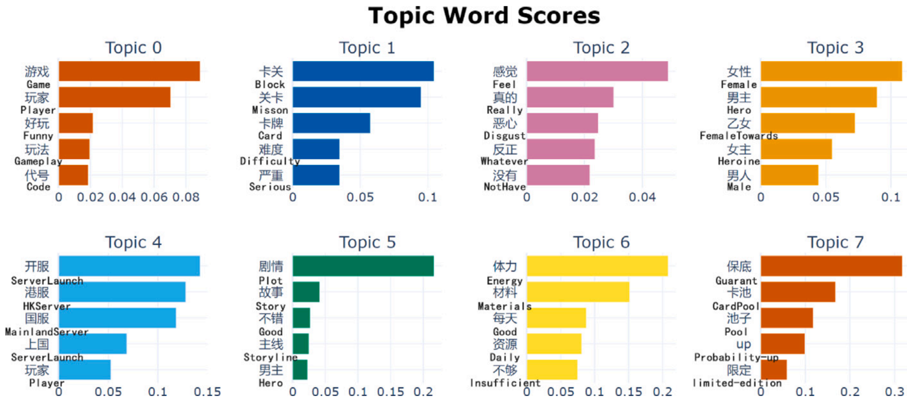

Fig. 5. Example of BERTopic.

the same category exhibit stronger personalized ranking requirements. Limiting the study to one category allows for a more consistent and meaningful evaluation of the proposed ranking method. To illustrate the personalized ranking process in a straightforward manner, we first present case studies of two customers as examples. Subsequently, large-scale comparative and ablation experiments are conducted on the entire dataset to rigorously evaluate the performance of the proposed method.

4.1. Dataset description

We used Python to scrape online reviews from Taptap (one of the largest video game forums in China, https://www.taptap.cn) for 8 popular romance games (P_1: Ashes of the Kingdom, P_2: Light and Night, P_3: The Moon in the Heart of Flowers and Mountains, P_4: Mr. Love: Queen's Choice, P_5: For All Time, P_6: Dream and Lethe Record, P_7: Tears of Themis, and P_8: Lady of Turbulent Era). A total of 72,000 reviews (9000 reviews for each game) were collected, contributed by 31,198 unique customers. The data scraped included customer ID, customer nickname, review time, review device, review score, and review text. After data preprocessing, we obtained 175,512 sub-sentences. From these, we randomly selected 10,000 samples for model training. A co-author and an experienced game industry professional manually labeled each sub-sentence as positive, negative, or neutral (the resulting Cohen's Kappa coefficient was 0.82), and any discrepancies were subsequently resolved through discussion to reach a final consensus. To evaluate the performance of the sentiment analysis model, 2000 samples were randomly selected as the test set, and the remaining 8000 were used as the training set.

4.2. Experiment results of numeralization of text data

After data preprocessing, we extracted the key CRs from the online reviews using the BERTopic model mentioned in Section 3.1. Then, we applied the BW-CNN model to identify the sentiment polarity of the reviews, thereby constructing the CR-sentiment matrix to convert the text data into numerical values.

4.2.1. CR extraction

First, we used the BERTopic model for topic modeling. A pre-trained bert_based_chinese model was used as the embedding model. In the application of HDBSCAN, we set the minimum cluster size option to 200 to ensure robust clustering results. The BERTopic model was instantiated with the following parameters: calculate_probabilities = True, min_topic_size = 'auto'. Additionally, we used an n-gram range of (1, 3) to extract the main feature words as output, while other parameters were set to their default values. And we obtained different numbers of topics for each of the 8 games. Fig. 5 shows, as an example for P_1, a bar chart illustrating the top 8 topics for P_1, with the key feature words of each topic determined by their c-TF-IDF scores.

Table 1
CRs in each game.

CRs	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
CR_1 (Story)	✓	✓	✓	✓	✓	✓	✓	✓
CR_2 (Resource)	✓		✓	✓		✓	✓	
CR_3 (Operation)	✓	✓	✓	✓	✓	✓	✓	
CR_4 (Character)	✓	✓	✓	✓	✓	✓	✓	✓
CR_5 (Payment)	✓	✓	✓	✓	✓	✓	✓	
CR_6 (Visual)	✓	✓	✓	✓	✓	✓	✓	✓
CR_7 (Sound)	✓	✓	✓	✓	✓	✓	✓	✓
CR_8 (Event)	✓	✓	✓	✓	✓	✓	✓	✓
CR_9 (Effort)	✓		✓	✓	✓	✓		
CR_{10} (Dialogue)	✓	✓			✓		✓	✓
CR_{11} (Mission)	✓			✓		✓		
CR_{12} (Romance)		✓		✓	✓		✓	
CR_{13} (Combat)			✓		✓	✓		
CR_{14} (Farming)			✓					

After being reviewed by experts, similar topics were consolidated into CRs, and irrelevant topics were discarded. The final CRs contained in the online reviews for each game are shown in Table 1.

Where the ✓symbol indicates that the corresponding CR appears in the respective game. CR_1 (Story), CR_3 (Operation), CR_4 (Character), CR_5 (Payment), CR_6 (Visual), CR_7 (Sound), and CR_8 (Event) are present in all eight games, highlighting their critical importance to the customer experience and suggesting they are key CRs influencing customers' game selection. CR_2 (Resource), CR_9 (Effort), and CR_{10} (Dialogue), though not present in every game, remain significant in most games. This indicates their notable influence on the customer experience. CR_2 (Resource) pertains to the management and utilization of in-game resources, which are crucial for shaping customers' strategies and game progression. CR_9 (Effort) focuses on the time and effort players invest in the game, which is essential for immersion and sustained customer interest. CR_{10} (Dialogue) addresses the quality of in-game dialogues, which enhance story depth and character interactions. These CRs are also widely recognized as basic CRs in games.

In contrast, CR_{11} (Mission), CR_{12} (Romance), CR_{13} (Combat), and CR_{14} (Farming) appear only in specific games, reflecting their connection to the core gameplay or design of those games. For example, games emphasizing combat mechanics are likely to prioritize CR_{13} (Combat), while games focusing on romantic simulation content may place greater importance on CR_{12} (Romance). These unique CRs reflect the distinct needs of customers in different games.

4.2.2. Sentiment analysis

We utilized a BW-CNN model employing a Double Embedding strategy, which leverages both BERT and Word2Vec to extract textual features at character and word levels. These two embeddings are then convolved to capture generalized and domain-specific features,

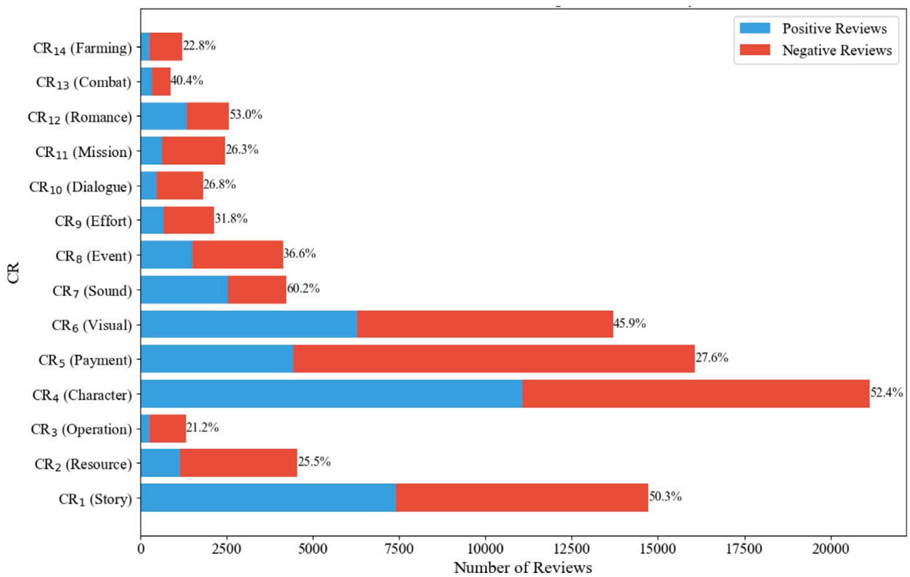

Fig. 6. Distribution of positive and negative reviews by CRs.

Table 2
Overall accuracy, Recall, and F1-score of seven models.

Model	Accuracy (%)	Recall	F1-score
BW-CNN	76.6	0.76	0.76
BW-CNN-Attention	74.1	0.74	0.73
Word2vec-CNN	72.1	0.71	0.72
BERT-CNN	72.3	0.70	0.72
LSTM	65.1	0.63	0.64
BiLSTM	70.8	0.69	0.70
GBDT	60.3	0.58	0.59

Table 3
Precision, Recall, and F1-score of BW-CNN for each sentiment category.

Emotion	Precision	Recall	F1-score
Positive	0.78	0.75	0.76
Neutral	0.74	0.70	0.72
Negative	0.80	0.84	0.82
Avg	**0.77**	**0.76**	**0.76**

Table 4
SEN_g^n of CRs.

CRs	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
CR_1 (Story)	0.33	0.41	0.39	0.52	0.60	0.60	0.40	0.44
CR_2 (Resource)	0.22	0.46	0.20	0.37	0.43	0.25	0.14	
CR_3 (Operation)	0.14		0.17	0.36		0.23	0.22	
CR_4 (Character)	0.39	0.50	0.35	0.64	0.64	0.50	0.42	0.54
CR_5 (Payment)	0.19	0.30	0.22	0.41	0.46	0.28	0.19	
CR_6 (Visual)	0.40	0.33	0.37	0.45	0.68	0.53	0.32	0.49
CR_7 (Sound)	0.48	0.53	0.49	0.65	0.65	0.67	0.42	0.54
CR_8 (Event)	0.22	0.29	0.23	0.42	0.50	0.46	0.25	0.26
CR_9 (Effort)	0.24		0.23	0.43	0.49	0.29		
CR_{10} (Dialogue)	0.24	0.25			0.47		0.17	0.20
CR_{11} (Mission)	0.25			0.36		0.37		
CR_{12} (Romance)		0.55		0.68	0.59		0.30	
CR_{13} (Combat)			0.27		0.46	0.56		
CR_{14} (Farming)			0.15			0.29		

enabling the model to identify the sentiment polarity of the text. In our model, the parameters were optimized through several experiments. The final filter heights were 2, 3, 4, and 5, $Learning\_Rate = 0.001$, $Dropout = 0.05$.

To evaluate the performance of the sentiment analysis model, we conducted seven experiments and assessed the model's accuracy on the test set. As shown in Table 2, the BW-CNN model achieved the highest overall accuracy (76.6%), recall (0.76), and F1-score (0.76) on our online review dataset. Among the tested models, we included a more complex variant by inserting a self-attention layer between the convolution and pooling layers to examine whether the relatively simple CNN architecture might limit performance. However, the attention-enhanced model slightly underperformed compared to BW-CNN, possibly due to overfitting or the limitations of the training data. These results suggest that the BW-CNN structure offers a favorable balance between effectiveness and robustness for sentiment classification in our context.

To further assess the model's effectiveness, especially in handling different sentiment types, we report the Precision, Recall, and F1-score for each model. These metrics, summarized in Table 3, provide a more detailed evaluation of model performance across positive, neutral, and negative categories.

Fig. 6 shows the total number of positive/negative reviews for each CR across all games. The percentage numbers in the bar chart represent the proportion of positive reviews. Overall, CR_1 (Story), CR_4 (Character), CR_5 (Payment), and CR_6 (Visual) received the highest number of reviews, indicating their prominence in customer discussions. Certain CRs, such as CR_1 (Story), CR_4 (Character), CR_7 (Sound), and CR_{12} (Romance), have a high proportion of positive reviews, all exceeding 50%, indicating that these CRs are highly appreciated by customers. However, CR_2 (Resource), CR_3 (Operation), CR_{10} (Dialogue), CR_{11} (Mission), and CR_{14} (Farming) have a lower proportion of positive reviews, below 30%, indicating that these CRs may have significant issues.

By the Eq. (12) in Section 3.3, the SEN_g^n of each CR in each game is calculated and shown in Table 4.

First, certain CRs such as CR_1 (Story), CR_4 (Character), CR_6 (Visual), and CR_7 (Sound) show consistently high proportions of positive reviews across multiple games. This indicates that these CRs are widely appreciated by customers across different game types, likely due to their critical role in enhancing game immersion and overall experience.

Second, while some CRs like CR_2 (Resource) and CR_9 (Effort) exhibit varied performance across games, they still achieve high positive review ratios in specific cases. For instance, CRs related to resource production and recycling received favorable evaluations in P_2 and P_5, demonstrating that these CRs are better aligned with the game design and player expectations. Overall, the data reveals the performance of various CRs across different games, providing valuable insights into customer satisfaction with different game CRs. These findings can

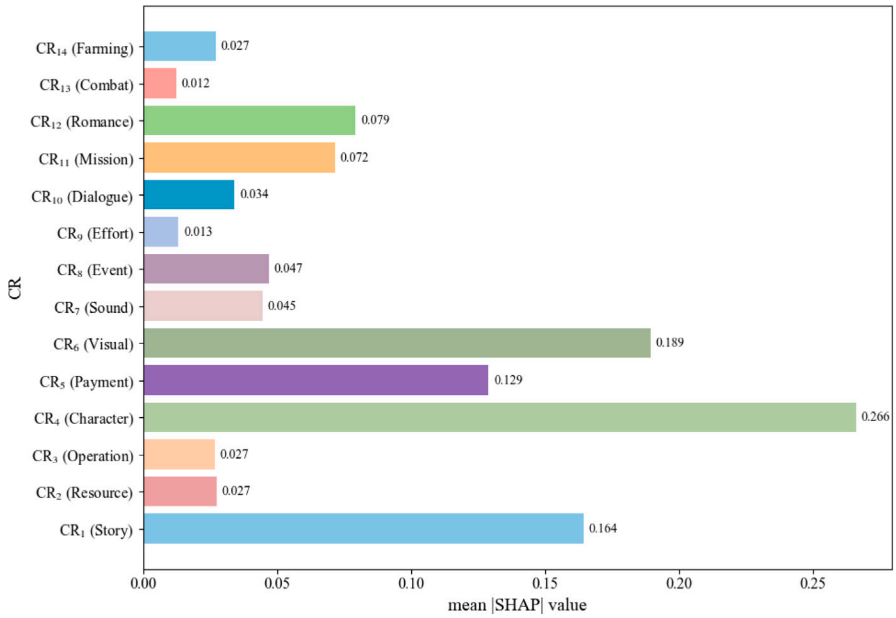

Fig. 7. Mean |SHAP| value of CRs.

inform future game design and improvements. However, satisfaction alone cannot directly infer the significant influence of a specific CR on overall customer satisfaction. Further data mining is necessary to explore these relationships.

4.3. Preference measurement

4.3.1. Measuring the influence of CRs based on SHAP

In this subsection, we train a Backpropagation Neural Network (BPNN) model and apply the SHAP method to interpret the model, thereby measuring the influence of customers' positive and negative sentiment expressions on their satisfaction with each CR. The BPNN consists of an input layer matching the feature dimension, two hidden layers with 64 and 32 neurons respectively, and an output layer producing a single continuous value representing the predicted rating. The ReLU activation function is used in the hidden layers. The model is optimized using the Adam optimizer, with a mean squared error (MSE) loss function. Training is performed for 200 epochs with a batch size of 32. SHAP values are then computed based on the trained BPNN to interpret the contribution of sentiment features to customer satisfaction.

The input features of the BPNN model are $\widetilde{POS}_{gk}^{n}$ and $\widetilde{NEG}_{gk}^{n}$ constructed in Section 3.1.4, with the target value being the RS_{gk}.

By SHAP method, we are able to quantify the influence of each CR's positive and negative sentiment on customer satisfaction. This method helps to understand how different CRs affect customers' overall satisfaction.

Based on Eqs. (5)–(7), we calculate the influence values of each CR in each game. The mean |SHAP| value of each CR in alternative games is shown in Fig. 7. CR_1 (Story), CR_4 (Character), CR_5 (Payment), and CR_6 (Visual) have higher mean |SHAP| values, indicating that the performance of these CRs has a significant influence on customer satisfaction. Moreover, these four CRs are also the ones with the highest number of reviews, which further reflects their high importance in the gaming experience.

The influence of each CR on each game is shown in Table 5. The positive sentiment influence of CR_4 (Character) is the highest among most games, especially in P_1 and P_8, with values of 0.420 and 0.039, respectively, demonstrating the significant positive contribution of character design to customer satisfaction. This suggests that character design quality plays a crucial role in enhancing overall customer satisfaction.

CR_5 (Payment) and CR_6 (Visual) stand out in terms of negative sentiment influence. Particularly, the negative influence values of CR5 in P_3 and P_7 are 0.207 and 0.165, respectively, indicating that customer dissatisfaction with in-game payment mechanisms and visual effects has a considerable negative influence on satisfaction.

The positive and negative sentiment influence values of CR_2 (Resource) and CR_9 (Effort) are relatively low, indicating a minimal influence of these CRs on customer satisfaction. The positive sentiment influence of CR_3 (Operation) is nearly zero, while its negative influence is relatively high in some games. This suggests that while customers have high expectations for this CR, any issues can significantly decrease satisfaction.

Combining the performance of CRs, certain high-performing CRs, such as CR_6 (Visual), also exhibit relatively high negative sentiment influence values. This indicates that although visual effects have received positive feedback in some games, there is also significant negative feedback. Meanwhile, low-performing CRs, such as CR_2 (Resource) and CR_9 (Effort), show low influence values, suggesting that despite receiving a large number of negative reviews, these CRs have not significantly influenced overall game satisfaction.

This highlights the limitations of analyzing CR performance alone and underscores the necessity of using interpretable machine learning methods to measure the influence of CRs on satisfaction.

4.3.2. Kano classification based on SHAP values

The median mean |SHAP| values, MED_{POS} and MED_{NEG}, are 0.0165 and 0.0235, respectively. This indicates that, overall, customers' negative sentiment has a greater influence on satisfaction than positive sentiment.

Based on SHAP values and the Kano classification rules defined in Eq. (8), the CRs in each game are categorized. The classification results are shown in Table 6. For better visualization, a scatter plot is also provided, as shown in Fig. 8.

In Fig. 8, the four regions are divided according to the values of MED_{POS} and MED_{NEG}. The CRs in the upper-left region are categorized as M-F CR, those in the upper-right region are O-F CR, those in the lower-right region are A-F CR, and those in the lower-left region are I-F CR.

Table 5
Influence of CRs.

	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
CR_1 (Story)-POS	0.059	0.057	0.119	0.045	0.047	0.110	0.133	0.169
CR_1 (Story)-NEG	0.074	0.069	0.018	0.032	0.148	0.071	0.013	0.151
CR_2 (Resource)-POS	0.028	0.015	0.037	0.001	0.013	0.001	0.017	0.001
CR_2 (Resource)-NEG	0.010	0.001	0.018	0.006	0.016	0.043	0.012	0.001
CR_3 (Operation)-POS	0.002	0.001	0.001	0.001	0.001	0.001	0.006	0.001
CR_3 (Operation)-NEG	0.062	0.001	0.030	0.018	0.001	0.068	0.023	0.001
CR_4 (Character)-POS	0.420	0.405	0.103	0.260	0.086	0.086	0.121	0.039
CR_4 (Character)-NEG	0.111	0.099	0.020	0.067	0.136	0.129	0.043	0.001
CR_5 (Payment)-POS	0.018	0.040	0.095	0.025	0.010	0.008	0.053	0.006
CR_5 (Payment)-NEG	0.142	0.039	0.207	0.104	0.087	0.165	0.025	0.006
CR_6 (Visual)-POS	0.091	0.157	0.113	0.031	0.034	0.102	0.031	0.292
CR_6 (Visual)-NEG	0.081	0.138	0.062	0.062	0.054	0.098	0.021	0.149
CR_7 (Sound)-POS	0.014	0.006	0.040	0.025	0.009	0.041	0.021	0.086
CR_7 (Sound)-NEG	0.002	0.008	0.002	0.013	0.019	0.041	0.004	0.024
CR_8 (Event)-POS	0.004	0.020	0.011	0.003	0.016	0.039	0.019	0.009
CR_8 (Event)-NEG	0.029	0.042	0.005	0.028	0.060	0.049	0.004	0.040
CR_9 (Effort)-POS	0.012	0.001	0.028	0.010	0.009	0.004	0.001	0.001
CR_9 (Effort)-NEG	0.006	0.001	0.003	0.011	0.001	0.017	0.001	0.001
CR_{10} (Dialogue)-POS	0.009	0.004	0.001	0.001	0.006	0.001	0.033	0.006
CR_{10} (Dialogue)-NEG	0.061	0.058	0.001	0.001	0.030	0.001	0.024	0.038
CR_{11} (Mission)-POS	0.146			0.002		0.002		
CR_{11} (Mission)-NEG	0.051			0.011		0.003		
CR_{12} (Romance)-POS		0.043		0.119	0.009		0.079	
CR_{12} (Romance)-NEG		0.012		0.015	0.010		0.030	
CR_{13} (Combat)-POS			0.004		0.003	0.007		
CR_{13} (Combat)-NEG			0.008		0.013	0.003		
CR_{14} (Farming)-POS			0.001			0.003		
CR_{14} (Farming)-NEG			0.010			0.041		

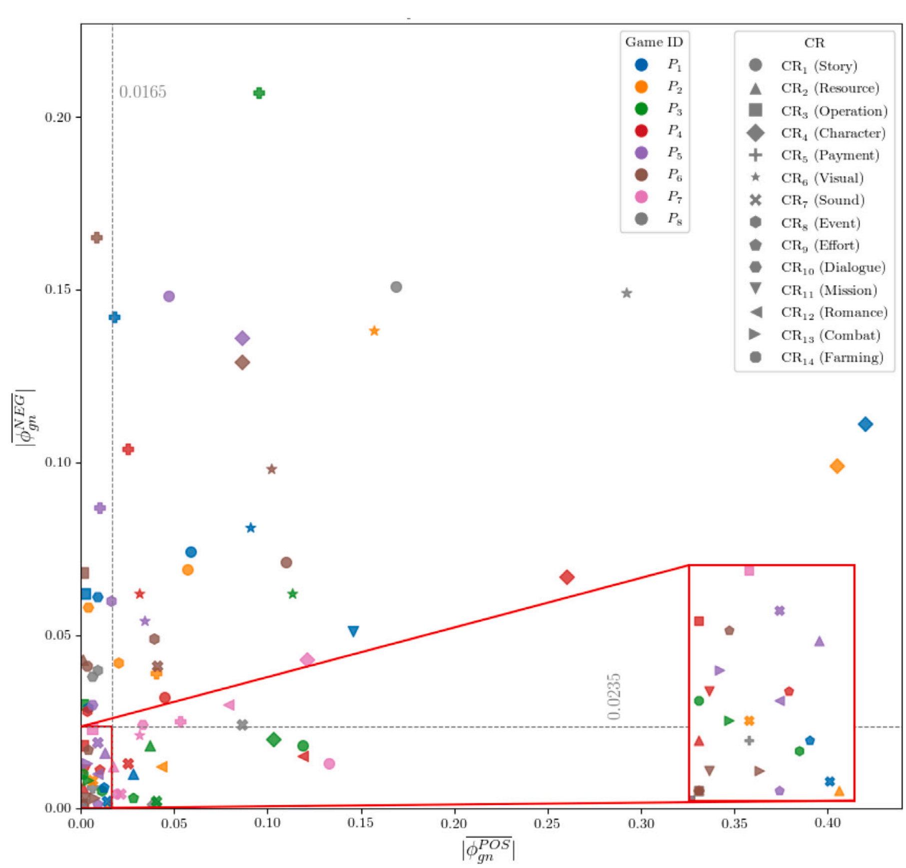

Fig. 8. Kano categories of CRs.

Table 6
Kano categories of CRs.

CRs	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
CR_1 (Story)	O-F	O-F	A-F	O-F	O-F	O-F	A-F	O-F
CR_2 (Resource)	A-F	I-F	A-F	I-F	I-F	M-F	A-F	I-F
CR_3 (Operation)	M-F	I-F	M-F	I-F	I-F	M-F	I-F	I-F
CR_4 (Character)	O-F	O-F	A-F	O-F	O-F	O-F	O-F	A-F
CR_5 (Payment)	O-F	O-F	O-F	O-F	M-F	M-F	O-F	I-F
CR_6 (Visual)	O-F	O-F	O-F	O-F	O-F	O-F	A-F	O-F
CR_7 (Sound)	I-F	I-F	A-F	A-F	I-F	O-F	A-F	O-F
CR_8 (Event)	M-F	O-F	I-F	M-F	M-F	O-F	A-F	M-F
CR_9 (Effort)	I-F	I-F	A-F	I-F	I-F	I-F	I-F	I-F
CR_{10} (Dialogue)	M-F	M-F	I-F	I-F	M-F	I-F	O-F	M-F
CR_{11} (Mission)	O-F			I-F		I-F		
CR_{12} (Romance)		A-F		A-F	I-F		O-F	
CR_{13} (Combat)			I-F		I-F	I-F		
CR_{14} (Farming)			I-F			M-F		

Table 7
Number of customers categorizing each CR into Kano categories.

CRs	O-F	A-F	M-F	I-F	Unmentioned
CR_1 (Story)	7404	939	62	45	21 100
CR_2 (Resource)	1983	944	256	97	26 270
CR_3 (Operation)	80	7	1103	11	28 349
CR_4 (Character)	11 432	468	40	3	17 607
CR_5 (Payment)	8190	34	562	4	20 760
CR_6 (Visual)	8750	171	44	2	20 583
CR_7 (Sound)	931	2387	1	6	26 225
CR_8 (Event)	2339	536	228	53	26 394
CR_9 (Effort)	56	613	449	742	27 690
CR_{10} (Dialogue)	385	51	948	5	28 161
CR_{11} (Mission)	219	311	964	8	28 048
CR_{12} (Romance)	1802	248	40	11	27 449
CR_{13} (Combat)	12	251	123	372	28 792
CR_{14} (Farming)	15	9	650	288	28 588

First, the I-F region contains more CRs, which are more densely distributed. This indicates that many CRs have a relatively small influence on customer satisfaction, and their importance is quite similar. Therefore, for these CRs, lower weights can be assigned in the ranking process, reducing their influence on the final ranking.

Second, the O-F region contains the most CRs, which are more loosely distributed. These CRs have a linear influence on customer satisfaction and play a significant role in customer reviews. The loose distribution suggests that, although all CRs in the O-F category share a similar influence type, their Imp_{gn} values differ significantly. This suggests that using only Kano categories to determine CR importance is insufficient; specific SHAP values must also be considered.

Finally, some CRs in the M-F and A-F regions have Imp_{gn} values close to those of the CRs in the O-F region. However, the high quality of the M-F CRs is the basic expectation of customers. On the other hand, A-F CRs are excitement factors: while their excellent performance significantly boosts customer satisfaction, their absence does not lead to dissatisfaction. Therefore, despite the similar Imp_{gn} values of these CRs, the different CR categories imply that their type of influence on customer satisfaction differs, and so does their importance. This highlights the necessity of distinguishing CR importance based on Kano categories to adjust customer preferences effectively.

Table 6 presents the Kano categories of each CR across different games. It reveals the categorization differences of each CR in various games, showing that the influence of the same CR on player satisfaction varies across games.

Some CRs exhibit more consistent categories across different games. For example, Story (CR1) is categorized as O-F CR in most games, indicating that this CR has a relatively consistent influence type: the better its performance, the higher the customer satisfaction. In contrast, some CRs show greater variation across different games. For instance, Resource (CR2) is classified as A-F CR in some games, where its high performance significantly enhances customer satisfaction, while in others, it is classified as I-F CR, indicating minimal influence on satisfaction. Similarly, Operation (CR3) is categorized as M-F CR in some games, suggesting that a decline in its performance significantly influences customer satisfaction, while in others, it is classified as I-F CR. This variation reflects differences in game characteristics and customer demographics.

These differences suggest that game developers need to tailor their strategies based on customer feedback specific to their games, determining the CR categories accordingly. This approach helps developers better identify and meet customer expectations, improving the game's appeal and satisfaction in a competitive market.

In addition to the cross-game comparison of Kano categories shown in Table 6, we further conducted a customer-level Kano classification analysis across all games. Table 7 summarizes the number of customers who classified each CR into different Kano categories: O-F (One-dimensional Feature), A-F (Attractive Feature), M-F (Must-be Feature), I-F (Indifferent Feature), and Unmentioned (customers who never referred to the CR in any review).

This table provides a more fine-grained and aggregated view of how customers perceive each CR across all games. Notably, CRs such as Story (CR1), Character (CR4), and Visual (CR6) have high proportions of customers categorizing them as O-F, reaffirming their consistent impact on satisfaction: better performance on these CRs directly leads to increased customer satisfaction.

In contrast, Operation (CR3) and Farming (CR14) exhibit higher proportions of M-F classifications, highlighting that customers are particularly sensitive to their deterioration, while improvements bring relatively little added value. Sound (CR7) shows a notably large share of A-F classifications, indicating that while not necessary, its excellence can significantly boost satisfaction. Some CRs like Effort (CR9) and Combat (CR13) display wide spreads across different categories, reflecting diverse customer expectations and experience orientations.

These findings reinforce the importance of personalized CR evaluation and the heterogeneity of player preferences. While game-level analysis reveals the overall trend of CR perceptions in specific titles, the customer-level Kano classification demonstrates that even within the same game, different customers assign different roles to the same CR. Therefore, a "one-size-fits-all" improvement or marketing strategy may fail to resonate with large segments of the player base.

This result validates the motivation of our personalized ranking method: by identifying which CRs are key drivers of satisfaction for which customers, developers and marketers can tailor their optimization or promotional strategies accordingly. Integrating these differentiated Kano roles with individualized importance metrics enables precise and effective decision-making for customer satisfaction management in video games.

4.4. Sample-based personalized ranking illustration

To provide an intuitive demonstration of how the proposed method captures individual-level heterogeneity, we select two customers, A (u = 551 494) and B (u = 34 869 6673), both of whom have posted detailed reviews for multiple games. Their personalized preference data are shown in Table 8.

As shown in Table 8, there exist considerable differences in both the preference strength and Kano category assigned to the same CR across different individuals. For instance, Customer A perceives CR5 (Payment) as O-F with relatively high positive SHAP contribution, whereas Customer B treats the same CR as M-F and expresses stronger negative sentiment toward it. This contrast suggests that customers may interpret and respond to the same CR in significantly different ways, influenced by their personal expectations, usage context, or prior experiences.

This illustrative example highlights the necessity of incorporating individual-level preference heterogeneity into the ranking model,

Table 8
Personalized preference data of Customer A and Customer B.

CR	Customer A (u = 551 94)			Customer B (u = 348 696 673)		
	$\lvert\phi_{un}^{POS}\rvert$	$\lvert\phi_{un}^{NEG}\rvert$	Kano	$\lvert\phi_{un}^{POS}\rvert$	$\lvert\phi_{un}^{NEG}\rvert$	Kano
CR_1 (Story)	0.11	0.03	O-F	0.02	0.04	O-F
CR_2 (Resource)		Unmentioned		0.05	0.01	A-F
CR_3 (Operation)	0.01	0.04	M-F	0.01	0.15	M-F
CR_4 (Character)	0.04	0.14	O-F	0.09	0.01	A-F
CR_5 (Payment)	0.10	0.04	O-F	0.01	0.14	M-F
CR_6 (Visual)	0.10	0.03	O-F	0.01	0.03	M-F
CR_7 (Sound)	0.01	0.01	I-F		Unmentioned	
CR_8 (Event)	0.04	0.03	O-F		Unmentioned	
CR_9 (Effort)		Unmentioned			Unmentioned	
CR_{10} (Dialogue)		Unmentioned			Unmentioned	
CR_{11} (Mission)	0.01	0.05	M-F		Unmentioned	
CR_{12} (Romance)	0.05	0.03	O-F	0.15	0.03	O-F
CR_{13} (Combat)		Unmentioned			Unmentioned	
CR_{14} (Farming)		Unmentioned			Unmentioned	
w_u^{M-F}		1.32			1.42	
w_u^{A-F}		Unmentioned			0.63	
w_u^{I-F}		0.68			0.21	

Table 9
Decision matrix after missing value imputation.

CRs	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
CR_1 (Story)	0.33	0.41	0.39	0.52	0.60	0.60	0.40	0.44
CR_2 (Resource)	0.22	0.46	0.20	0.37	0.43	0.25	0.14	0.30
CR_3 (Operation)	0.14	0.22	0.17	0.36	0.22	0.23	0.22	0.22
CR_4 (Character)	0.39	0.50	0.35	0.64	0.64	0.50	0.42	0.54
CR_5 (Payment)	0.19	0.30	0.22	0.41	0.46	0.28	0.19	0.50
CR_6 (Visual)	0.40	0.33	0.37	0.45	0.68	0.53	0.32	0.49
CR_7 (Sound)	0.48	0.53	0.49	0.65	0.65	0.67	0.42	0.54
CR_8 (Event)	0.22	0.29	0.23	0.42	0.50	0.46	0.25	0.26
CR_9 (Effort)	0.24	0.34	0.23	0.43	0.49	0.29	0.33	0.34
CR_{10} (Dialogue)	0.24	0.25	0.27	0.27	0.47	0.27	0.17	0.20
CR_{11} (Mission)	0.25	0.25	0.25	0.36	0.25	0.37	0.25	0.25
CR_{12} (Romance)	0.30	0.55	0.30	0.68	0.59	0.30	0.30	0.30
CR_{13} (Combat)	0.27	0.27	0.27	0.27	0.46	0.56	0.27	0.27
CR_{14} (Farming)	0.15	0.15	0.15	0.15	0.15	0.29	0.15	0.15

Table 10
Rating scores and the ranking calculated by our method.

Game	Customer A (u = 551 494)		Customer B (u = 348 696 673)	
	Rating score	TOPSIS rank	Rating score	TOPSIS rank
P_1	2	8	–	8
P_2	3	5	1	3
P_3	4	7	–	7
P_4	5	2	4	1
P_5	–	1	4	2
P_6	–	3	–	4
P_7	–	6	1	6
P_8	–	4	–	5
Spearman ρ	0.80		0.89	

rather than relying solely on population-level averages. While the following subsection reports systematic comparative experiments across the entire dataset, this sample-based demonstration offers a concrete view of the personalized ranking process.

The decision matrix after missing value imputation is shown in Table 9.

Fig. 9 shows the similarity of customer preferences across different games. Most games exhibit high similarity in customer preferences, suggesting they likely cater to similar customer groups or operate in comparable market environments. On the other hand, the low similarity between certain games highlights significant differences in customer preferences. This suggests that these games have distinct market positioning or target different customer groups.

The coefficient α reflects the customer's sensitivity to public opinion. In this study, $\alpha = 0.2$ was selected by systematically searching the interval $[0, 1]$ with a step size of 0.1, in order to achieve an optimal balance between individual preferences and public opinion. The robustness of this choice is further examined in Section 4.6, where different values of α are tested to evaluate their impact on the ranking results.

The rating score provided by the two customers and the ranking calculated by our method are shown in Table 10.

As shown in Table 10, the personalized ranking results for Customers A and B illustrate how our method reflects individual heterogeneity in preferences. For Customer A, P_4 received the highest rating score of 5 and was ranked second in the personalized ranking, while P_3 and P_1, which were rated relatively lower (scores of 4 and 2, respectively), were ranked seventh and eighth. Interestingly, P_5—which was not explicitly rated by Customer A—appeared at the top of the personalized ranking, indicating that the model can generate predictions even for games without direct rating information. For Customer B, both P_4 and P_5 were given the highest score of 4 and were ranked first and second, respectively, in the personalized results, while games with lower scores (e.g., P_2, P_6, P_7) were assigned correspondingly lower ranks overall.

These cases are intended as illustrative demonstrations of how the method operates in practice, rather than as statistical validation of its accuracy. The potential discrepancies between observed scores and predicted rankings (e.g., the position of P_5 for Customer A or P_2 for Customer B) highlight the complexity of capturing latent preferences, but do not by themselves establish model superiority. Rigorous evidence is provided in the subsequent large-scale comparison and ablation experiments, where statistical evaluation metrics such as Spearman correlation are systematically reported.

4.5. Large-scale comparison of methods

To analyze the role of each component in our ranking method, we conducted a series of ablation experiments targeting Kano importance, public opinion, and personalized preferences. For clarity, we explicitly define each variant as follows:

(1) **DK** Kano categories are excluded by setting all Kano weights w_{gn} to 1.
(2) **DPO** Public opinion adjustment is excluded by setting the coefficient $\alpha = 0$.
(3) **DPP** Personalized preferences are excluded by computing the global average of all customers' R_{un}^{imp}, which is then used as the unified preference representation for ranking across all alternatives.

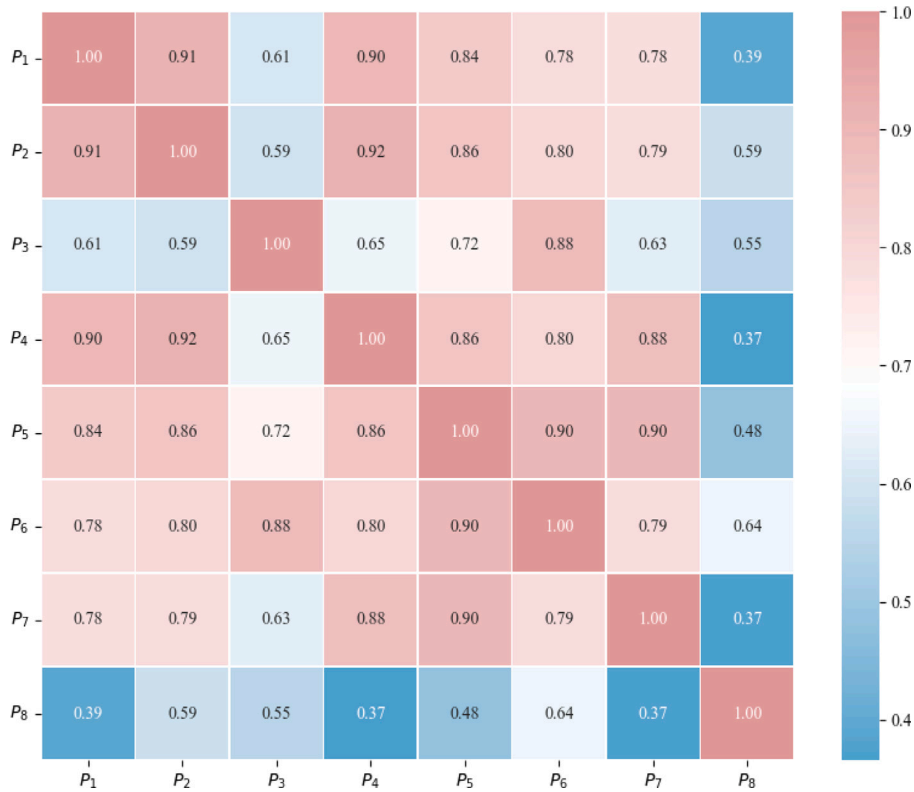

Fig. 9. Similarity of customer preferences.

To systematically evaluate the contribution of each component, we design ablation variants by excluding different combinations of Kano adjustment (DK), public opinion adjustment (DPO), and personalized preferences (DPP). Specifically:

(1) **DK** Kano categories are excluded by setting all Kano weights w_{gn} to 1.
(2) **DPO** Public opinion adjustment is excluded by setting the coefficient $\alpha = 0$.
(3) **DPP** Personalized preferences are excluded by computing the global average of all customers' R_{un}^{imp}, which is then used as the unified preference representation for ranking across all alternatives.

Based on these three ablation rules, we construct the following variants:

- **Method-Deletion1 (DK)**: Only Kano categories are excluded.
- **Method-Deletion2 (DPO)**: Only public opinion is excluded.
- **Method-Deletion3 (DPP)**: Only personalized preferences are excluded.
- **Method-Deletion4 (DK+DPO)**: Both Kano categories and public opinion are excluded.
- **Method-Deletion5 (DK+DPP)**: Both Kano categories and personalized preferences are excluded.
- **Method-Deletion6 (DPO+DPP)**: Both public opinion and personalized preferences are excluded.
- **Method-Deletion7 (Original TOPSIS)(DK+DPO+DPP)**: All three components are excluded, producing a unified ranking solely based on unadjusted customer group preferences.

To validate the effectiveness of the proposed method, we compare it with two existing MCDM-based ranking methods: Method1 [72] and Method2 [73]. Method1 employs information entropy for attribute weighting and utilizes regret theory for ranking. Method2 uses attribute frequency for weighting and implements an improved PageRank algorithm for ranking. In the comparative experiment, to ensure data consistency, numerical values from text data are standardized using the proposed method.

In addition, we include a baseline method that ranks games purely based on customers' average rating scores, referred to as "Score-Based Ranking". This baseline provides a straightforward yet intuitive benchmark for evaluating the alignment between customers' explicit ratings and the computed rankings. And to further benchmark personalized ranking performance, we additionally include a collaborative filtering (CF) approach based on Singular Value Decomposition (SVD), which infers user preferences from historical rating patterns.

To assess the consistency between the computed rankings and individual customer preferences, we calculate the Spearman rank correlation coefficient between each customer's review score sequence (including only customers who rated at least three games) and the ranking results obtained by each method. This evaluation reflects how well each method aligns with the personalized preferences implicitly revealed through the customers' own rating behaviors.

We then take the average Spearman correlation across all qualifying customers as an overall metric for each method's ranking consistency from an individual perspective.

Fig. 10 shows that the proposed method achieves the highest average Spearman correlation (0.452), confirming that jointly integrating Kano importance, public opinion, and personalized preferences produces rankings that are best aligned with individual customer preferences.

The ablation variants all perform worse than the full method. Among the single-deletion cases, removing personalized preferences (Method-Deletion 3 and 6, 0.417) results in the largest performance drop, highlighting the crucial role of individualized preference modeling. Removing Kano categories (Method-Deletion 1, 0.438) or public opinion adjustment (Method-Deletion 2, 0.442) also reduces performance, though to a lesser degree.

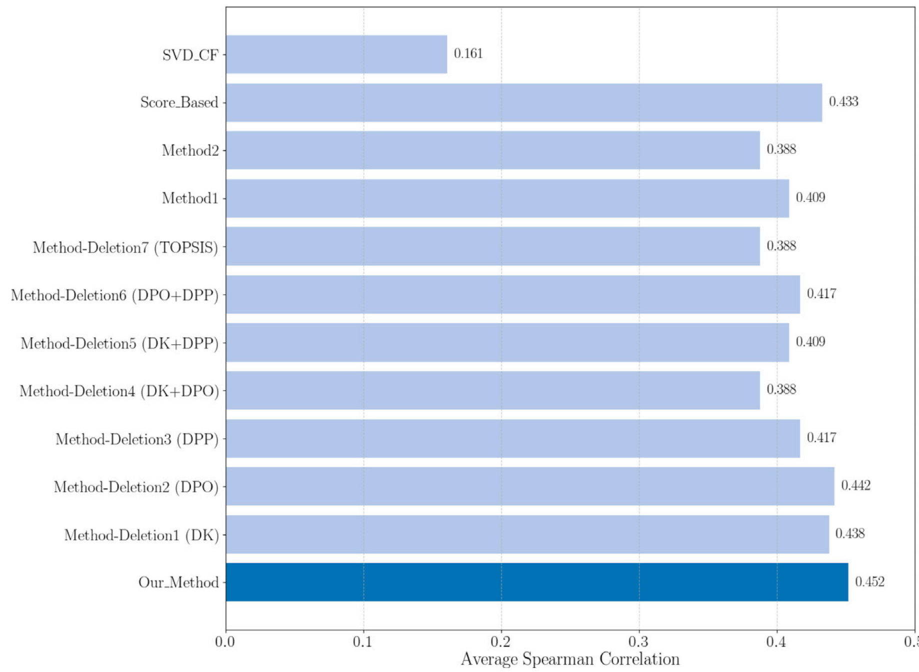

Fig. 10. Comparison of ranking methods.

Some ablation variants show identical average Spearman correlation (e.g., Method-Deletion 3 and 6 both 0.417). This occurs because, when personalized preferences are excluded, the ranking is generated using a single unified preference vector computed across all customers. Consequently, even though different components (Kano, public opinion) are adjusted differently, the overall effect on the unified ranking may be the same, resulting in identical correlation scores.

When two components are simultaneously removed, the degradation becomes more pronounced. Method-Deletion 4 (DK+DPO, 0.388) and Method-Deletion 7 (TOPSIS baseline without enhancements, 0.388) achieve the lowest results, indicating that without Kano and public opinion adjustments the framework loses much of its explanatory power. Method-Deletion 5 (DK+DPP, 0.409) and Method-Deletion 6 (DPO+DPP, 0.417) also perform notably worse than the complete method, showing that personalized preferences are indispensable regardless of whether Kano or public opinion information is included.

In comparison, Method1 (0.409) and Method2 (0.388) show results similar to the weaker ablation variants, underscoring that conventional MCDM methods cannot capture the same degree of personalization achieved by the proposed framework. The Score-Based method (0.433) achieves a relatively high correlation, which is expected since the evaluation metric is derived from user rating sequences. Nevertheless, it fails to account for heterogeneous user preferences, allowing the proposed method to outperform it. By contrast, the SVD-based collaborative filtering approach performs the worst (0.161), suggesting that matrix factorization alone is insufficient in this setting, likely due to data sparsity and the lack of explicit requirement-level interpretability.

Addition, Since SVD-based CF generates user-specific rankings, we introduce two widely adopted top-k evaluation metrics – Precision@3 and NDCG@3 – to assess the relevance and ranking quality of the recommendations. Specifically, Precision@3 measures the proportion of relevant items among the top 3 recommendations for each user, and is defined as Eq. (20)

$$\text{Precision@3}_u = \frac{|\text{Top}_3(u) \cap \text{Rel}(u)|}{3} \tag{20}$$

where $\text{Top}_3(u)$ denotes the top-3 recommended items and $\text{Rel}(u)$ denotes the set of relevant items for user u, defined as items with truth review scores ≥ 3.0.

Table 11
Performance comparison between our method and collaborative filtering (SVD).

Method	Spearman ρ	Precision@3	NDCG@3
Our method	0.452	0.244	0.506
SVD-based CF	0.161	0.185	0.415

NDCG@3 evaluates not only the presence of relevant items but also their ranking positions, and is given by Eq. (21)

$$\text{NDCG@3}_u = \frac{1}{\text{IDCG@3}_u} \sum_{i=1}^{3} \frac{2^{rel_i} - 1}{\log_2(i+1)} \tag{21}$$

where $rel_i = 1$ if the item at position i is relevant, and 0 otherwise. The Ideal DCG (IDCG@3) represents the maximum possible DCG score that can be obtained if all relevant items are ranked in the highest positions. It serves as a reference to evaluate how close the actual ranking is to this ideal case.

Likewise, we restrict the evaluation to users who have rated at least three different games to ensure the validity of the ranking metrics.

As shown in Table 11, our method significantly outperforms the SVD-based CF approach across all three evaluation metrics, demonstrating superior overall ranking consistency and top-3 recommendation quality.

4.6. Sensitivity analysis

We perform a sensitivity analysis on the parameter α, which controls the influence of public opinion.

From Table 12, we observe that the average Spearman correlation initially increases with α, reaching a peak at $\alpha = 0.2$, and then gradually decreases as α continues to grow. This pattern suggests that although the improvement is modest, incorporating public opinion from other games helps simulate the influence of external views on customer preferences, leading to a more accurate personalized ranking.

When α is too low, the ranking primarily relies on public opinion derived from games that the user has personally played. This may neglect valuable signals from similar games that the user has not yet

Table 12
Sensitivity analysis of α.

α	Avg. spearman	α	Avg. spearman
0.0	0.442	0.6	0.435
0.1	0.446	0.7	0.432
0.2	**0.452**	0.8	0.436
0.3	0.443	0.9	0.435
0.4	0.437	1.0	0.434
0.5	0.437		

experienced. Conversely, when α becomes too high, the influence of opinions from other games dominates, which may introduce noise or dilute the relevance of feedback tied directly to the user's own experience.

5. Discussion

This section focuses on the broader implications and limitations of the proposed personalized ranking method. On the one hand, the theoretical and practical implications are discussed. On the other hand, we acknowledge several limitations of the study, which provide promising directions for future research.

5.1. Research implications

From a theoretical perspective, to the best of our knowledge, the personalized ranking method proposed in this study is the first to systematically integrate Kano's requirement typology with SHAP-based explainability techniques for modeling user preferences in the domain of video games. While previous research has applied requirement classification methods [74] or explainable machine learning [75] approaches in ranking methods, our study uniquely bridges these areas to produce a more psychologically grounded and interpretable preference modeling framework. This integration offers a new direction for research at the intersection of human-centric product design and explainable AI.

Another theoretical contribution of this study lies in the proposed personalized ranking framework that models individual-level preferences. Unlike prior approaches that typically treat all users as homogeneous or rely on collaborative signals alone, our method constructs a unique preference profile for each user by aggregating SHAP across multiple reviews they have provided. This method captures the nuanced ways in which different users value distinct requirement categories and supports the construction of personalized rankings that closely reflect each user's specific evaluative priorities. And we introduce a novel adjustment mechanism that accounts for public opinion influence in scenarios. This approach acknowledges the real-world context in which customers are not isolated evaluators but are often influenced by the broader reputation landscape shaped by peer reviews and digital word-of-mouth.

From a practical perspective, this study not only proposes a more personalized ranking method but also provides a quantitative approach to uncovering the specific CRs that users care about in this kind of video games. Our framework enables the extraction and measurement of CR-level user concerns, even when such concerns are only implicitly expressed in online reviews. This capability is especially valuable for game developers seeking to align product features with user expectations. Our framework outputs highlight that some CRs – such as Character, Story, Visual, or Payment mechanics – are the most influential in shaping user evaluations. Such insights can guide targeted game improvements, inform CR prioritization decisions, and support the design of customer-centered updates that address the true drivers of customer satisfaction. Ultimately, the practical value of our method lies in its dual capability to empower customers with personalized choices while simultaneously equipping developers with interpretable insights into customer expectations—bridging the gap between customer experience and product evolution in the highly competitive video game industry.

From a computational perspective, the proposed method is efficient and feasible for practical use. All experiments were conducted on a system with an Intel i5-9400 CPU, NVIDIA RTX 2060 GPU, and 16 GB of RAM. On our dataset, the BERTopic model took approximately 5 min to run, the BW-CNN model around 15 min, the SHAP around 2 min, and the personalized ranking process about 1 min. While runtime may vary depending on hardware conditions or system load, these results indicate that the overall computational cost is modest. Therefore, the method is suitable for deployment in real-world scenarios and can scale to moderate-size game recommendation platforms.

5.2. Limitations

While our proposed method demonstrates performance in aligning rankings with individual customer preferences, several limitations should be acknowledged.

First, our proposed framework is methodologically generalizable and could be applied to other platforms or game genres. However, the empirical findings in this study are based solely on the Taptap platform and focus only on romance/otome games. The dataset is limited in scope. It does not include other genres, indie titles, or games from different release periods. Moreover, the demographic characteristics and review culture of Taptap users may differ from those of other platforms such as Steam or Google Play. As a result, the observed patterns – such as user preferences, requirement categorization, and sentiment dynamics – may not directly generalize to other user communities or game types. Future work should consider a more diverse and heterogeneous dataset to improve robustness and generalizability.

Second, while we employ interpretable and computationally efficient models such as BW-CNN for sentiment analysis and a BPNN for preference learning, we acknowledge that more advanced models may offer superior performance. However, such models typically require larger labeled datasets and more computational resources, which were beyond the scope of this study. Future research may explore the trade-off between interpretability and predictive power using state-of-the-art architectures.

Third, the Kano classification process in this study adopts a fixed-threshold approach for categorizing customer requirements. While this method offers clarity, it may oversimplify the nuanced nature of user perceptions. Alternative methods that incorporate probabilistic or data-driven thresholds could potentially yield more flexible and robust categorizations.

Finally, the current preference modeling framework relies solely on review text and does not incorporate behavioral signals such as playtime, purchase history, or clickstream data. These behavioral features, which are widely adopted in practical recommendation systems, could significantly enhance the completeness and accuracy of user profiles. In future work, integrating such behavioral data with text-based analysis will be an important direction to improve both the robustness and the applicability of our framework. Until then, our method can be regarded as a complementary module for text-driven personalization, particularly in scenarios where behavioral data is unavailable or restricted.

6. Conclusions

With the rapid expansion of the video game market, delivering accurate personalized game rankings has become a critical research focus. This study addresses key limitations in existing ranking methods, including challenges in capturing personalized preferences, the lack of consideration for variations in CR category importance, and the absence of a mechanism to incorporate insights from public opinion. To overcome these issues, we propose an S-Kano-TOPSIS ranking method for video games, integrating requirement categories and public opinion.

This method unifies personalized preference measurement, CR category classification, and personalized ranking into a cohesive framework. Compared to traditional approaches, our method provides deeper insights into personalized preferences, improves ranking precision by adjusting preference weights based on CR categories, and incorporates relevant public opinion to reflect real-world decision contexts. Empirical results based on reviews from eight romance/otome games confirm the effectiveness of our approach: the proposed method improves Spearman correlation by 0.064 over the baseline TOPSIS, and outperforms a CF approach by 0.30 in Spearman correlation, 0.06 in Precision@3, and 0.09 in NDCG@3. These results validate the performance advantages of the proposed method and demonstrate its potential for applications in personalized video game ranking.

Specifically, CRs are extracted using the BERTopic model, and sentiment analysis is conducted with the BW-CNN model to assess their performance. The SHAP method is applied to quantify the influence of each CR on customer satisfaction, serving as the basis for measuring personalized preferences. The Kano model is then used to classify CRs based on their influence modes, and the importance of each CR is adjusted accordingly. Additionally, since customers often consider multiple similar games when making purchasing decisions, the preference measurement is further refined by incorporating relevant public opinion. Finally, the S-Kano-TOPSIS method is applied to generate a personalized ranking that aligns with the customer's actual needs. An empirical study based on online review data from eight similar video games demonstrates that the proposed method effectively captures personalized preferences and provides tailored rankings, helping customers select games that best match their expectations. This approach holds promise for broader applications beyond the video game industry.

Despite its effectiveness, the proposed method has certain limitations. First, it heavily relies on online review data, which may be influenced by data acquisition channels and does not incorporate behavioral metrics such as playtime. Second, the Kano classification process follows predefined rules, which may require further optimization to improve the accuracy of CR importance measurement. Moreover, the current dataset is limited to romance/otome games collected from a single platform, which may restrict the generalizability of the findings to other game genres or platforms with different user demographics and review cultures. Future research could integrate additional user behavior data to enrich personalized ranking dimensions, expand the dataset across diverse game categories and platforms, and refine the model's performance.

CRediT authorship contribution statement

Yanze Liu: Writing – original draft, Visualization, Software, Methodology. **Tian-Hui You:** Writing – review & editing, Validation, Supervision, Project administration. **Junrong Zou:** Writing – review & editing, Validation. **Yuan Yuan:** Writing – review & editing, Validation, Supervision. **Bing-Bing Cao:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Data sources

The online review data used in this study were collected from the following websites:

- P_1: Ashes of the Kingdom: https://www.taptap.cn/app/239090/review
- P_2: Light and Night: https://www.taptap.cn/app/186316/review
- P_3: The Moon in the Heart of Flowers and Mountains: https://www.taptap.cn/app/195161/review
- P_4: Mr. Love: Queen's Choice: https://www.taptap.cn/app/55307/review
- P_5: For All Time: https://www.taptap.cn/app/173797/review
- P_6: Dream and Lethe Record: https://www.taptap.cn/app/187661/review
- P_7: Tears of Themi: https://www.taptap.cn/app/172095/review
- P_8: Alkaid: Lady of Turbulent Era: https://www.taptap.cn/app/226664/review

Appendix B. Model hyperparameters

To ensure reproducibility of the experiments, we provide detailed hyperparameter settings for all key models used in this study.

B.1. Bertopic parameters

- Embedding model: `bert-based-chinese`
- Vectorization method: SentenceTransformers
- Dimensionality reduction: UMAP
  - `n_neighbors` = 15
  - `n_components` = 5
  - `metric` = `cosine`
- Clustering algorithm: HDBSCAN
  - `min_cluster_size` = 200
  - `min_samples` = 1
  - `cluster_selection_method` = '`eom`'
- `min_topic_size` = 'auto'
- n-gram range: (1, 3)
- Number of top keywords per topic: 10
- `diversity` = 0.3 (to improve topic distinctiveness)
- `calculate_probabilities` = True

B.2. BW-CNN parameters

- Input representation:
  - Dual-channel input: BERT + Word2Vec
  - BERT model:
    - Embedding model: `bert-based-chinese`
    - Embedding dimension: 768
  - Word2Vec embeddings:
    - Embedding: 150-dim, skip-gram, window = 3, min_count = 10, epochs = 200
    - Vocabulary size: 6616 (from corpus)
- Convolutional layers:
  - Filter sizes: [2, 3, 4, 5]
  - Number of filters per size: 100

- Activation function: ReLU

- Pooling layers:
  - Max-over-time pooling
- Fully connected layers:
  - Activation function: Softmax
- Optimization:
  - Loss function: CrossEntropyLoss
  - Optimizer: Adam
  - Dropout rate: 0.05
  - Learning rate: 0.001
  - Batch size: 128
  - Epochs: Trained until no improvement in validation accuracy for 500 consecutive epochs (early stopping); best model saved.
- Evaluation metric: Accuracy, Precision, Recall, F1-score (macro-averaged)

B.3. BPNN parameters

- Hidden layers: 2
- Hidden units: [64, 32]
- Activation function: ReLU
- Loss function: Mean Squared Error (MSE)
- Optimizer: Adam
- Learning rate: 0.001
- Batch size: 32
- Epochs: 200

B.4. SHAP settings

- Explainer types: KernelExplainer

Appendix C. Data preprocessing and splits

C.1. Data preprocessing

- Tokenization: Using Jieba tokenizer with a customized dictionary.
- Segmentation: Split into sub-sentences, but segments shorter than 5 Chinese characters are kept intact to preserve key phrases.
- Stopwords removal: A domain-specific stopword list is constructed by extending the HowNet stopword list with proprietary terms relevant to the video game industry. Words matching any entry in this stopword list are removed from the reviews through exact string matching using Python.
- Text cleaning: Removing non-Chinese characters, punctuation, and special symbols using Python.

C.2. Data splits

The dataset was randomly split into training and testing sets using Python's scikit-learn library with an 80-20 ratio. Specifically, the train-test-split function was applied with a fixed random seed to ensure reproducibility. The training set contains 80% of the samples for model training, while the remaining 20% serves as the testing set for evaluation. This stratified random split preserves the distribution of labels across both subsets.

Data availability

Data will be made available on request.

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